

EstunRemoteApiLib

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Chapter 1

命名空间索引

1.1 命名空间列表

这里列出了所有命名空间定义,附带简要说明:

EstunApiForPublic		7
EstunApiStruct_CLI		7

Chapter 2

类索引

2.1 类列表

这里列出了所有类、结构、联合以及接口定义等，并附带简要说明:

EstunApiStruct_CLI::E_ROB_ADDDO_IO_CLI	13
EstunApiStruct_CLI::E_ROB_APIBUFF_CLI	15
EstunApiStruct_CLI::E_ROB_COORD_CLI	16
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EstunApiForPublic::EstunApi_CLI	28

Chapter 3

文件索引

3.1 文件列表

这里列出了所有文件，并附带简要说明:

EstunApiForPublic.h	79
EstunApiStruct_CLI.h	84

Chapter 4

命名空间文档

4.1 EstunApiForPublic 命名空间参考

类

- class [EstunApi_CLI](#)

4.2 EstunApiStruct_CLI 命名空间参考

类

- struct [E_ROB_ADDDDO_IO_CLI](#)
- struct [E_ROB_APIBUFF_CLI](#)
- struct [E_ROB_COORD_CLI](#)
- struct [E_ROB_JOBID_CLI](#)
- struct [E_ROB_PAYLOAD_CLI](#)
- struct [E_ROB_PERMIT_CLI](#)
- struct [E_ROB_POS_CLI](#)
- struct [E_ROB_PROGNAME_CLI](#)
- struct [E_ROB_SPEED_CLI](#)
- struct [E_ROB_ZONE_CLI](#)

枚举

- enum class [E_IOType_CLI](#) {
 DI = 0 , DO = 1 , AO = 2 , AI = 3 ,
 SIM_DI = 11 , SIM_DO = 12 , SIM_AI = 13 , SIM_AO = 14 }
- enum class [E_VarType_CLI](#) {
 OtherType = 0 , intType = 1 , realType = 2 , aposType = 3 ,
 cposType = 4 , strType = 5 , RealIntArray = 6 , tooType = 7 ,
 userType = 8 , CposArray = 11 , AposArray = 12 }
- enum class [E_ScopeType_CLI](#) { system = 0 , global , project , local }
- enum class [E_CoordType_CLI](#) {
 axisCoord = 0 , worldCoord , toolCoord , usrCoord ,
 errCoord }

- enum class `E_UserCoordType_CLI` { `UsrCoord` = 0 , `ExtCoord` }
- enum class `E_DistentionType_CLI` { `negative` = 0 , `forward` }
- enum class `E_SysModeType_CLI` {
 `manualMode` = 0 , `autoMode` , `remoteMode` , `banMode` ,
 `mutil` , `API` , `errSysMode` }
- enum class `E_RunModeType_CLI` { `step` = 1 , `sustain` , `errRunMode` }
- enum class `E_RunStatusType_CLI` { `running` = 1 , `suspend` , `stop` , `errRunStatus` }
- enum class `E_ServoStatusType_CLI` { `ServoOff` = 0 , `ServoOn` , `errStatus` }

4.2.1 枚举类型说明

4.2.1.1 E_CoordType_CLI

```
enum class EstunApiStruct_CLI::E_CoordType_CLI [strong]
```

坐标类型 0-关节坐标, 1-世界坐标, 2-工具坐标, 3-用户坐标, 4-错误

枚举值

axisCoord	
worldCoord	
toolCoord	
usrCoord	
errCoord	

在文件 `EstunApiStruct_CLI.h` 第 58 行定义.

4.2.1.2 E_DistentionType_CLI

```
enum class EstunApiStruct_CLI::E_DistentionType_CLI [strong]
```

方向类型 1-为正方向, 0-负方向

枚举值

negative	
forward	

在文件 `EstunApiStruct_CLI.h` 第 81 行定义.

4.2.1.3 E_IOType_CLI

```
enum class EstunApiStruct_CLI::E_IOType_CLI [strong]
```


IO类型. 0-DI, 1-DO, 2-AO, 3-AI, 11-SIM_DI, 12-SIM_DO, 13-SIM_AI, 14-SIM_AO

枚举值

DI	
DO	
AO	
AI	
SIM_DI	
SIM_DO	
SIM_AI	
SIM_AO	

在文件 [EstunApiStruct_CLI.h](#) 第 11 行定义.

4.2.1.4 E_RunModeType_CLI

```
enum class EstunApiStruct_CLI::E_RunModeType_CLI [strong]
```

运动模式类型 1-单步, 2-连续, 3-错误

枚举值

step	
sustain	
errRunMode	

在文件 [EstunApiStruct_CLI.h](#) 第 106 行定义.

4.2.1.5 E_RunStatusType_CLI

```
enum class EstunApiStruct_CLI::E_RunStatusType_CLI [strong]
```

运动状态类型. 1-运行, 2-暂停, 3-停止, 4-错误

枚举值

running	
suspend	
stop	
errRunStatus	

在文件 [EstunApiStruct_CLI.h](#) 第 117 行定义.

4.2.1.6 E_ScopeType_CLI

```
enum class EstunApiStruct_CLI::E_ScopeType_CLI [strong]
```

变量域类型. 0-系统, 1-全局, 2-工程, 3-程序本地

枚举值

system	
global	
project	
local	

在文件 [EstunApiStruct_CLI.h](#) 第 46 行定义.

4.2.1.7 E_ServoStatusType_CLI

```
enum class EstunApiStruct_CLI::E_ServoStatusType_CLI [strong]
```

mot使能状态. 0-未使能, 1-使能, 3-错误

枚举值

ServoOff	
ServoOn	
errStatus	

在文件 [EstunApiStruct_CLI.h](#) 第 129 行定义.

4.2.1.8 E_SysModeType_CLI

```
enum class EstunApiStruct_CLI::E_SysModeType_CLI [strong]
```

系统模式类型 0-手动, 1-自动, 2-远程, 3-禁止, 4-多工位, 5-API, 6-错误

枚举值

manualMode	
autoMode	
remoteMode	
banMode	
mutil	
API	
errSysMode	

在文件 [EstunApiStruct_CLI.h](#) 第 91 行定义.

4.2.1.9 E_UserCoordType_CLI

```
enum class EstunApiStruct_CLI::E_UserCoordType_CLI [strong]
```

用户坐标系类型 0-用户坐标, 1-外部坐标系

枚举值

UsrCoord	
ExtCoord	

在文件 [EstunApiStruct_CLI.h](#) 第 71 行定义.

4.2.1.10 E_VarType_CLI

```
enum class EstunApiStruct_CLI::E_VarType_CLI [strong]
```

数据类型. 0-其他类型, 1-整型, 2-实数, 3-关节位置, 4-空间位置, 5-字符串, 6-实数数组, 7-工具参数, 8-用户坐标参数, 11-空间坐标数组, 12-轴关节坐标数组

枚举值

OtherType	
intType	
realType	
aposType	
cposType	
strType	
RealIntArray	
tooType	
userType	
CposArray	
AposArray	

在文件 [EstunApiStruct_CLI.h](#) 第 27 行定义.

Chapter 5

类说明

5.1 EstunApiStruct_CLI::E_ROB_ADDDO_IO_CLI结构体 参考

```
#include <EstunApiStruct_CLI.h>
```

Public 属性

- [E_IOType_CLI ioType](#)
- [System::Int32 ioindex](#)
- [System::Double iovalue](#)
- [System::Int32 trigType](#)
- [System::Int32 exTimes](#)
- [System::Int32 exPercent](#)
- [System::Double exDistance](#)

5.1.1 详细描述

附加指令结构一.

在文件 [EstunApiStruct_CLI.h](#) 第 212 行定义.

5.1.2 类成员变量说明

5.1.2.1 exDistance

```
System::Double EstunApiStruct_CLI::E_ROB_ADDDO_IO_CLI::exDistance
```

当前运动执行触发距离.

在文件 [EstunApiStruct_CLI.h](#) 第 241 行定义.

5.1.2.2 exPercent

```
System::Int32 EstunApiStruct_CLI::E_ROB_ADDDO_IO_CLI::exPercent
```

当前运动执行触发百分比.

在文件 [EstunApiStruct_CLI.h](#) 第 237 行定义.

5.1.2.3 exTimes

```
System::Int32 EstunApiStruct_CLI::E_ROB_ADDDO_IO_CLI::exTimes
```

触发延时,单位毫秒, 负数为提前触发.

在文件 [EstunApiStruct_CLI.h](#) 第 233 行定义.

5.1.2.4 ioindex

```
System::Int32 EstunApiStruct_CLI::E_ROB_ADDDO_IO_CLI::ioindex
```

IO序号.

在文件 [EstunApiStruct_CLI.h](#) 第 221 行定义.

5.1.2.5 ioType

```
E\_IOType\_CLI EstunApiStruct_CLI::E_ROB_ADDDO_IO_CLI::ioType
```

IO类型.

在文件 [EstunApiStruct_CLI.h](#) 第 217 行定义.

5.1.2.6 iovalue

```
System::Double EstunApiStruct_CLI::E_ROB_ADDDO_IO_CLI::iovalue
```

IO值.

在文件 [EstunApiStruct_CLI.h](#) 第 225 行定义.

5.1.2.7 trigType

System::Int32 EstunApiStruct_CLI::E_ROB_ADDDO_IO_CLI::trigType

触发类型（1:路径百分比；2: 距离）。

在文件 [EstunApiStruct_CLI.h](#) 第 229 行定义。

该结构体的文档由以下文件生成:

- [EstunApiStruct_CLI.h](#)

5.2 EstunApiStruct_CLI::E_ROB_APIBUFF_CLI结构体 参考

```
#include <EstunApiStruct_CLI.h>
```

Public 成员函数

- [E_ROB_APIBUFF_CLI\(\)](#)

Public 属性

- System::SByte [m_type](#)
- System::Collections::Generic::List< System::SByte > ^ [m_data](#)

5.2.1 详细描述

附加指令，具体内容由其他结构参数转换得到，需要调用AddDoTypeConvert相关接口。

在文件 [EstunApiStruct_CLI.h](#) 第 188 行定义。

5.2.2 构造及析构函数说明

5.2.2.1 E_ROB_APIBUFF_CLI()

```
EstunApiStruct_CLI::E_ROB_APIBUFF_CLI::E_ROB_APIBUFF_CLI ( ) [inline]
```

构造函数。

在文件 [EstunApiStruct_CLI.h](#) 第 194 行定义。

5.2.3 类成员变量说明

5.2.3.1 m_data

```
System::Collections::Generic::List<System::SByte> ^ EstunApiStruct_CLI::E_ROB_APIBUFF_CLI::m_data
```

指令内容二进制.

在文件 [EstunApiStruct_CLI.h](#) 第 206 行定义.

5.2.3.2 m_type

```
System::SByte EstunApiStruct_CLI::E_ROB_APIBUFF_CLI::m_type
```

附加指令类型.

在文件 [EstunApiStruct_CLI.h](#) 第 202 行定义.

该结构体的文档由以下文件生成:

- [EstunApiStruct_CLI.h](#)

5.3 EstunApiStruct_CLI::E_ROB_COORD_CLI结构体 参考

```
#include <EstunApiStruct_CLI.h>
```

Public 属性

- System::Double [x](#)
- System::Double [y](#)
- System::Double [z](#)
- System::Double [a](#)
- System::Double [b](#)
- System::Double [c](#)

5.3.1 详细描述

用户坐标.

在文件 [EstunApiStruct_CLI.h](#) 第 373 行定义.

5.3.2 类成员变量说明

5.3.2.1 a

System::Double EstunApiStruct_CLI::E_ROB_COORD_CLI::a

相对于世界（基）坐标系 z 轴旋转的欧拉角，单位是 deg.

在文件 [EstunApiStruct_CLI.h](#) 第 390 行定义.

5.3.2.2 b

System::Double EstunApiStruct_CLI::E_ROB_COORD_CLI::b

相对于世界（基）坐标系 y 轴旋转的欧拉角，单位是 deg.

在文件 [EstunApiStruct_CLI.h](#) 第 394 行定义.

5.3.2.3 c

System::Double EstunApiStruct_CLI::E_ROB_COORD_CLI::c

TCP 相对于世界（基）坐标系 x 轴旋转的欧拉角，单位是 deg.

在文件 [EstunApiStruct_CLI.h](#) 第 398 行定义.

5.3.2.4 x

System::Double EstunApiStruct_CLI::E_ROB_COORD_CLI::x

TCP 相对于世界（基）坐标系在 x 方向的位移偏移量，单位是 mm.

在文件 [EstunApiStruct_CLI.h](#) 第 378 行定义.

5.3.2.5 y

System::Double EstunApiStruct_CLI::E_ROB_COORD_CLI::y

TCP 相对于世界（基）坐标系在 y 方向的位移偏移量，单位是 mm.

在文件 [EstunApiStruct_CLI.h](#) 第 382 行定义.

5.3.2.6 z

```
System::Double EstunApiStruct_CLI::E_ROB_COORD_CLI::z
```

TCP 相对于世界（基）坐标系在 z 方向的位移偏移量，单位是 mm.

在文件 [EstunApiStruct_CLI.h](#) 第 386 行定义.

该结构体的文档由以下文件生成:

- [EstunApiStruct_CLI.h](#)

5.4 EstunApiStruct_CLI::E_ROB_JOBID_CLI结构体 参考

```
#include <EstunApiStruct_CLI.h>
```

Public 属性

- System::Int32 [m_jobID](#)
- System::UInt32 [m_timeStamp](#)

5.4.1 详细描述

指令标识信息.

在文件 [EstunApiStruct_CLI.h](#) 第 154 行定义.

5.4.2 类成员变量说明

5.4.2.1 m_jobID

```
System::Int32 EstunApiStruct_CLI::E_ROB_JOBID_CLI::m_jobID
```

大于0表示指令下发成功，否则表示指令下发失败.

在文件 [EstunApiStruct_CLI.h](#) 第 159 行定义.

5.4.2.2 m_timeStamp

System::UInt32 EstunApiStruct_CLI::E_ROB_JOBID_CLI::m_timeStamp

运动指令下发成功的当前时间戳.

在文件 [EstunApiStruct_CLI.h](#) 第 163 行定义.

该结构体的文档由以下文件生成:

- [EstunApiStruct_CLI.h](#)

5.5 EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI结构体 参考

```
#include <EstunApiStruct_CLI.h>
```

Public 属性

- System::Double [m_M](#)
- System::Double [m_Mx](#)
- System::Double [m_My](#)
- System::Double [m_Mz](#)
- System::Double [m_Ixx](#)
- System::Double [m_Iyy](#)
- System::Double [m_Izz](#)
- System::Double [m_Ixy](#)
- System::Double [m_Ixz](#)
- System::Double [m_Iyz](#)

5.5.1 详细描述

机器人负载.

在文件 [EstunApiStruct_CLI.h](#) 第 326 行定义.

5.5.2 类成员变量说明

5.5.2.1 m_Ixx

System::Double EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI::m_Ixx

安装的工具或装夹的负载在重心位的 X 方向回转的惯量矩, 单位是 kg mm².

在文件 [EstunApiStruct_CLI.h](#) 第 347 行定义.

5.5.2.2 m_lxy

```
System::Double EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI::m_lxy
```

安装的工具或装夹的负载在重心位的 XY 交叉方向的惯量积，单位是 kg mm².

在文件 [EstunApiStruct_CLI.h](#) 第 359 行定义.

5.5.2.3 m_lxz

```
System::Double EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI::m_lxz
```

安装的工具或装夹的负载在重心位的 XZ 交叉方向的惯量积，单位是 kg mm².

在文件 [EstunApiStruct_CLI.h](#) 第 363 行定义.

5.5.2.4 m_lyy

```
System::Double EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI::m_lyy
```

安装的工具或装夹的负载在重心位的 Y 方向回转的惯量矩，单位是 kg mm².

在文件 [EstunApiStruct_CLI.h](#) 第 351 行定义.

5.5.2.5 m_lyz

```
System::Double EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI::m_lyz
```

安装的工具或装夹的负载在重心位的 YZ 交叉方向的惯量积，单位是 kg mm².

在文件 [EstunApiStruct_CLI.h](#) 第 367 行定义.

5.5.2.6 m_lzz

```
System::Double EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI::m_lzz
```

安装的工具或装夹的负载在重心位的 Z 方向回转的惯量矩，单位是 kg mm².

在文件 [EstunApiStruct_CLI.h](#) 第 355 行定义.

5.5.2.7 m_M

System::Double EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI::m_M

工具&负载的重量， 单位是 kg.

在文件 [EstunApiStruct_CLI.h](#) 第 331 行定义.

5.5.2.8 m_Mx

System::Double EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI::m_Mx

安装的工具或装夹的负载在坐标系 OTool-XYZ 的 X 方向上的偏移量， 单位是 mm.

在文件 [EstunApiStruct_CLI.h](#) 第 335 行定义.

5.5.2.9 m_My

System::Double EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI::m_My

安装的工具或装夹的负载在坐标系 OTool-XYZ 的 Y 方向上的偏移量， 单位是 mm.

在文件 [EstunApiStruct_CLI.h](#) 第 339 行定义.

5.5.2.10 m_Mz

System::Double EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI::m_Mz

安装的工具或装夹的负载在坐标系 OTool-XYZ 的 Z 方向上的偏移量， 单位是 mm.

在文件 [EstunApiStruct_CLI.h](#) 第 343 行定义.

该结构体的文档由以下文件生成:

- [EstunApiStruct_CLI.h](#)

5.6 EstunApiStruct_CLI::E_ROB_PERMIT_CLI结构体 参考

```
#include <EstunApiStruct_CLI.h>
```

Public 属性

- System::UInt16 [m_mainctrlcode](#)
- System::UInt16 [m_timestamp](#)

5.6.1 详细描述

权限信息.

在文件 [EstunApiStruct_CLI.h](#) 第 139 行定义.

5.6.2 类成员变量说明

5.6.2.1 m_mainctrlcode

```
System::UInt16 EstunApiStruct_CLI::E_ROB_PERMIT_CLI::m_mainctrlcode
```

权限唯一码，大于1的整数表示有运动操作权限，否则没有.

在文件 [EstunApiStruct_CLI.h](#) 第 144 行定义.

5.6.2.2 m_timestamp

```
System::UInt16 EstunApiStruct_CLI::E_ROB_PERMIT_CLI::m_timestamp
```

时间戳，时间戳大于0表示正确获取到权限，否则获取权限失败.

在文件 [EstunApiStruct_CLI.h](#) 第 148 行定义.

该结构体的文档由以下文件生成:

- [EstunApiStruct_CLI.h](#)

5.7 EstunApiStruct_CLI::E_ROB_POS_CLI结构体 参考

```
#include <EstunApiStruct_CLI.h>
```

Public 成员函数

- [E_ROB_POS_CLI \(\)](#)

Public 属性

- System::Int32 [posType](#)
- System::Collections::Generic::List< System::Int32 > ^ [posCfg](#)
- System::Collections::Generic::List< System::Double > ^ [posValue](#)

5.7.1 详细描述

机器人位置.

在文件 [EstunApiStruct_CLI.h](#) 第 247 行定义.

5.7.2 构造及析构函数说明

5.7.2.1 E_ROB_POS_CLI()

```
EstunApiStruct_CLI::E_ROB_POS_CLI::E_ROB_POS_CLI ( ) [inline]
```

构造函数.

在文件 [EstunApiStruct_CLI.h](#) 第 253 行定义.

5.7.3 类成员变量说明

5.7.3.1 posCfg

```
System::Collections::Generic::List<System::Int32> ^ EstunApiStruct_CLI::E_ROB_POS_CLI::posCfg
```

元素个数7, CPos下生效, 空间位置配置.

在文件 [EstunApiStruct_CLI.h](#) 第 266 行定义.

5.7.3.2 posType

```
System::Int32 EstunApiStruct_CLI::E_ROB_POS_CLI::posType
```

位置类型, 0-APos, 1-CPos.

在文件 [EstunApiStruct_CLI.h](#) 第 262 行定义.

5.7.3.3 posValue

```
System::Collections::Generic::List<System::Double> ^ EstunApiStruct_CLI::E_ROB_POS_CLI::pos↵
Value
```

元素个数16, APos下为各轴关节坐标{j1,j2,j3...j16}, Cpos下为空间坐标值和附加轴坐标{x,y,z,a,b,c,a7...a16}.

在文件 [EstunApiStruct_CLI.h](#) 第 270 行定义.

该结构体的文档由以下文件生成:

- [EstunApiStruct_CLI.h](#)

5.8 EstunApiStruct_CLI::E_ROB_PROGNAME_CLI结构体 参考

```
#include <EstunApiStruct_CLI.h>
```

Public 属性

- System::String ^ [prjname](#)
- System::String ^ [programname](#)
- System::String ^ [exeprogramName](#)

5.8.1 详细描述

工程程序名.

在文件 [EstunApiStruct_CLI.h](#) 第 169 行定义.

5.8.2 类成员变量说明

5.8.2.1 exeprogramName

```
System::String ^ EstunApiStruct_CLI::E_ROB_PROGNAME_CLI::exeprogramName
```

当前运行的程序名.

在文件 [EstunApiStruct_CLI.h](#) 第 182 行定义.

5.8.2.2 prjname

```
System::String ^ EstunApiStruct_CLI::E_ROB_PROGNAME_CLI::prjname
```

当前加载的工程名.

在文件 [EstunApiStruct_CLI.h](#) 第 174 行定义.

5.8.2.3 programname

```
System::String ^ EstunApiStruct_CLI::E_ROB_PROGNAME_CLI::programname
```

当前加载的程序名.

在文件 [EstunApiStruct_CLI.h](#) 第 178 行定义.

该结构体的文档由以下文件生成:

- [EstunApiStruct_CLI.h](#)

5.9 EstunApiStruct_CLI::E_ROB_SPEED_CLI结构体 参考

```
#include <EstunApiStruct_CLI.h>
```

Public 属性

- System::Double [m_Per](#)
- System::Double [m_Tcp](#)
- System::Double [m_ori](#)
- System::Double [m_exj_l](#)
- System::Double [m_exj_r](#)

5.9.1 详细描述

机器人速度.

在文件 [EstunApiStruct_CLI.h](#) 第 276 行定义.

5.9.2 类成员变量说明

5.9.2.1 m_exj_l

```
System::Double EstunApiStruct_CLI::E_ROB_SPEED_CLI::m_exj_l
```

外部轴线速度.

在文件 [EstunApiStruct_CLI.h](#) 第 293 行定义.

5.9.2.2 m_exj_r

```
System::Double EstunApiStruct_CLI::E_ROB_SPEED_CLI::m_exj_r
```

外部轴角速度.

在文件 [EstunApiStruct_CLI.h](#) 第 297 行定义.

5.9.2.3 m_ori

```
System::Double EstunApiStruct_CLI::E_ROB_SPEED_CLI::m_ori
```

空间旋转速度.

在文件 [EstunApiStruct_CLI.h](#) 第 289 行定义.

5.9.2.4 m_Per

```
System::Double EstunApiStruct_CLI::E_ROB_SPEED_CLI::m_Per
```

关节速度百分比.

在文件 [EstunApiStruct_CLI.h](#) 第 281 行定义.

5.9.2.5 m_Tcp

```
System::Double EstunApiStruct_CLI::E_ROB_SPEED_CLI::m_Tcp
```

TCP 线速度.

在文件 [EstunApiStruct_CLI.h](#) 第 285 行定义.

该结构体的文档由以下文件生成:

- [EstunApiStruct_CLI.h](#)

5.10 EstunApiStruct_CLI::E_ROB_ZONE_CLI结构体 参考

```
#include <EstunApiStruct_CLI.h>
```

Public 属性

- System::Int32 [m_ZoneType](#)
- System::Double [m_Per](#)
- System::Double [m_dis](#)
- System::Int32 [m_vConst](#)

5.10.1 详细描述

机器人转弯区.

在文件 [EstunApiStruct_CLI.h](#) 第 303 行定义.

5.10.2 类成员变量说明

5.10.2.1 m_dis

```
System::Double EstunApiStruct_CLI::E_ROB_ZONE_CLI::m_dis
```

笛卡尔空间转弯区大小.

在文件 [EstunApiStruct_CLI.h](#) 第 316 行定义.

5.10.2.2 m_Per

```
System::Double EstunApiStruct_CLI::E_ROB_ZONE_CLI::m_Per
```

关节轨迹转弯百分比.

在文件 [EstunApiStruct_CLI.h](#) 第 312 行定义.

5.10.2.3 m_vConst

```
System::Int32 EstunApiStruct_CLI::E_ROB_ZONE_CLI::m_vConst
```

恒速过渡使能状态.

在文件 [EstunApiStruct_CLI.h](#) 第 320 行定义.

5.10.2.4 m_ZoneType

System::Int32 EstunApiStruct_CLI::E_ROB_ZONE_CLI::m_ZoneType

过渡类型(0-无过渡/1-绝对过渡/2-相对过渡).

在文件 [EstunApiStruct_CLI.h](#) 第 308 行定义.

该结构体的文档由以下文件生成:

- [EstunApiStruct_CLI.h](#)

5.11 EstunApiForPublic::EstunApi_CLI类 参考

```
#include <EstunApiForPublic.h>
```

Public 成员函数

- [EstunApi_CLI](#) ([System::Runtime::InteropServices::Optional] System::String^ configPath)
- [~EstunApi_CLI](#) ()
- System::Void [setLogConfig](#) (System::String^ logDir, System::Int32 size)
- System::Void [setLogEnable](#) (System::Boolean enable)
- System::Void [writeLog](#) (System::String^ str)
- System::String ^ [getLastErr](#) ()
- System::Boolean [connectToServer](#) (System::String^ ipAddr, System::Boolean isAutoReConnect)
- System::Boolean [disconnect](#) ()
- System::Boolean [isConnected](#) ()
- System::Int32 [E_GetAPIStatus](#) ()
- [E_ROB_PERMIT_CLI](#) ^ [E_AcquirePermit](#) ()
- [E_ROB_PERMIT_CLI](#) ^ [E_CurrentPermit](#) ()
- System::Boolean [E_ReleasePermit](#) ([E_ROB_PERMIT_CLI](#)^ permit)
- System::Boolean [E_MotionStart](#) ()
- System::Boolean [E_MotionPause](#) ()
- System::Boolean [E_MotionContinue](#) ()
- System::Boolean [E_MotionStop](#) ()
- System::Int32 [E_GetRobotRunStatus](#) ()
- [E_ROB_APIBUFF_CLI](#) ^ [ADDDO.IO.2.E_ROB_APIBUFF](#) ([E_ROB_ADDDO.IO_CLI](#)^ from)
- [E_ROB_JOBID_CLI](#) ^ [E_MovJ](#) (System::String^ Pname, [System::Runtime::InteropServices::Optional] System::String^ velName, [System::Runtime::InteropServices::Optional] System::String^ blendType, [System::Runtime::InteropServices::Optional] System::String^ zname, [System::Runtime::InteropServices::Optional] System::String^ tname, [System::Runtime::InteropServices::Optional] System::String^ coordname, [System::Runtime::InteropServices::Optional] System::String^ payloadName, [System::Runtime::InteropServices::Optional] [E_ROB_APIBUFF_CLI](#)^ addDo, [System::Runtime::InteropServices::Optional] System::Double valfine, [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish)
- [E_ROB_JOBID_CLI](#) ^ [E_MovJ2](#) ([E_ROB_POS_CLI](#)^ dstPoint, [System::Runtime::InteropServices::Optional] [E_ROB_SPEED_CLI](#)^ velo, [System::Runtime::InteropServices::Optional] [E_ROB_ZONE_CLI](#)^ zone, [System::Runtime::InteropServices::Optional] System::Int32 payloadId, [System::Runtime::InteropServices::Optional] System::Int32 toolId, [System::Runtime::InteropServices::Optional] System::Int32 coordId, [System::Runtime::InteropServices::Optional] [E_ROB_APIBUFF_CLI](#)^ addDo, [System::Runtime::InteropServices::Optional] System::Double valfine, [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish)

- [illegible]

- `E_ROB_JOBID_CLI ^ E_MovLArch2 (E_ROB_POS_CLI^ dstPoint, [System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo, [System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone, [System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ connerZone, [System::Runtime::InteropServices::Optional] System::Int32 payloadId, [System::Runtime::InteropServices::Optional] System::Int32 toolId, [System::Runtime::InteropServices::Optional] System::Int32 coordId, [System::Runtime::InteropServices::Optional] System::Double Upheight, [System::Runtime::InteropServices::Optional] System::Double Downheight, [System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo, [System::Runtime::InteropServices::Optional] System::Double valfine, [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish)`
- `System::Boolean E_GetToolsID (System::Int32% num, System::Collections::Generic::List< System::Int32 >^% idArray)`
- `System::Boolean E_CreateTool (System::Int32 toolID, E_ROB_COORD_CLI^ toolCoord, E_ROB_PAYLOAD_CLI^ payloadvalue)`
- `System::Boolean E_ModifyTool (System::Int32 toolID, E_ROB_COORD_CLI^ toolCoord, E_ROB_PAYLOAD_CLI^ payloadvalue)`
- `System::Boolean E_LoadTool (System::Int32 toolID)`
- `System::Int32 E_GetCurToolID ()`
- `System::String ^ E_GetToolName (System::Int32 toolID)`
- `System::Boolean E_GetTool (System::Int32 toolID, E_ROB_COORD_CLI^% toolCoord, E_ROB_PAYLOAD_CLI^% toolPayload)`
- `System::Boolean E_GetUserCoordsID (System::Int32% num, System::Collections::Generic::List< System::Int32 >^% idArray)`
- `System::Boolean E_CreateUserCoord (System::Int32 usrID, E_ROB_COORD_CLI^ usrData)`
- `System::Boolean E_ModifyUserCoord (System::Int32 usrID, E_ROB_COORD_CLI^ usrData)`
- `System::Boolean E_LoadUserCoord (System::Int32 usrID)`
- `System::Int32 E_GetCurUserCoordID (E_UserCoordType_CLI% coordtype)`
- `System::String ^ E_GetUserCoordName (System::Int32 coordID, E_UserCoordType_CLI coordtype)`
- `System::Boolean E_GetUserCoord (System::Int32 coordID, E_UserCoordType_CLI coordtype, E_ROB_COORD_CLI^% coord)`
- `System::Boolean E_GetPayLoadsID (System::Int32% num, System::Collections::Generic::List< System::Int32 >^% idArray)`
- `System::Boolean E_CreatePayLoad (System::Int32 payloadID, E_ROB_PAYLOAD_CLI^ payloadData)`
- `System::Boolean E_ModifyPayLoad (System::Int32 payloadID, E_ROB_PAYLOAD_CLI^ payloadData)`
- `System::Boolean E_LoadPayLoad (System::Int32 payloadID)`
- `System::Int32 E_GetCurPayLoadID ()`
- `System::String ^ E_GetPayLoadName (System::Int32 payloadID)`
- `System::Boolean E_GetPayLoad (System::Int32 payloadID, E_ROB_PAYLOAD_CLI^% payload)`
- `System::Boolean E_GetDefaultPayload (System::Int32% payloadID, System::String^% name, E_ROB_PAYLOAD_CLI^% payload)`
- `System::Boolean E_SetIOValue (System::Int32 IoIndex, E_IOTYPE_CLI IoType, System::Collections::Generic::List< System::Double >^% dataIn)`
- `System::Boolean E_GetIOValue (System::Int32 IoIndex, E_IOTYPE_CLI IoType, System::Int32 IoNum, System::Collections::Generic::List< System::Double >^% dataOut)`
- `System::Boolean E_SetVarValue (E_VarType_CLI varType, System::String^ varName, System::String^ varValue, E_ScopeType_CLI varScope)`
- `System::Boolean E_GetVarValue (E_VarType_CLI varType, System::String^ varName, E_ScopeType_CLI varScope, System::String^% valueBack)`
- `System::Boolean E_SetFine (System::Double value)`
- `System::Boolean E_GetFine (System::Double % value)`
- `System::Boolean E_SetJogCoord (E_CoordType_CLI coordType)`
- `E_CoordType_CLI E_GetJogCoord ()`
- `System::Boolean E_SetJogEnable (System::Boolean isEnabled)`
- `System::Boolean E_GetJogEnable ()`
- `System::Boolean E_JogMotion (System::Int32 axisId, E_DistentionType_CLI distention, System::Boolean mov)`

- System::Boolean [E_JogMicroMotion](#) (System::Int32 axisId, [E_DistentionType_CLI](#) distention, System::Double dist, System::Boolean mov)
- System::Boolean [E_TeachPoint](#) (System::String^ pointName, [E_ScopeType_CLI](#) scope)
- System::Boolean [E_GetRobotType](#) (System::String^% value)
- System::Boolean [E_GetRobotInfo](#) (System::String^% value)
- System::Int32 [E_GetRobotStatus](#) ()
- System::Boolean [E_SetGlobalSpeed](#) (System::Int32 value)
- System::Int32 [E_GetGlobalSpeed](#) ()
- System::Boolean [E_SetSysMode](#) ([E_SysModeType_CLI](#) sysMode)
- [E_SysModeType_CLI](#) [E_GetSysMode](#) ()
- System::Boolean [E_SetRunMode](#) ([E_RunModeType_CLI](#) mode)
- [E_RunModeType_CLI](#) [E_GetCurRunMode](#) ()
- System::Boolean [E_SetServoOn](#) (System::Boolean isOn)
- [E_ServoStatusType_CLI](#) [E_GetServoOn](#) ()
- System::Boolean [E_CheckClearError](#) ()
- System::Boolean [E_ClearError](#) ()
- System::Int32 [E_GetErrorId](#) ()
- System::String ^ [E_GetErrorString](#) (System::Int32 errorId)
- System::String ^ [E_GetErrorString](#) (System::Int32 errorId, System::String^ language)
- [E_ROB_POS_CLI](#) ^ [E_GetCurJPos](#) ()
- [E_ROB_POS_CLI](#) ^ [E_GetCurWPos](#) ()
- System::Boolean [E_SetExternalTCPByName](#) (System::String^ ecplName)
- System::Boolean [E_SetExternalTCPById](#) (System::Int32 ecplid)
- System::Boolean [E_GetCurProgramName](#) ([E_ROB_PROGNAME_CLI](#)^ % value)
- [E_RunStatusType_CLI](#) [E_GetCurProgramRunStatus](#) ()
- System::Boolean [E_UnLoadProject](#) ()
- System::Boolean [E_UnLoadProgram](#) ()
- System::Boolean [E_LoadProject](#) (System::String^ prjname)
- System::Boolean [E_LoadProgame](#) (System::String^ prjname, System::String^ programname)
- System::Boolean [E_ProgRun](#) ()
- System::Boolean [E_ProgPause](#) ()
- System::Boolean [E_ProgStop](#) ()
- System::Boolean [E_ReadInt](#) (System::String^ varName, System::Int32 varScope, System::Int32 % value↵ Back)
- System::Boolean [E_WriteInt](#) (System::String^ varName, System::Int32 varScope, System::Int32 value)
- System::Boolean [E_ReadReal](#) (System::String^ varName, System::Int32 varScope, System::Double % valueBack)
- System::Boolean [E_WriteReal](#) (System::String^ varName, System::Int32 varScope, System::Double value)
- System::Boolean [E_ReadStr](#) (System::String^ varName, System::Int32 varScope, System::String^% value↵ Back)
- System::Boolean [E_WriteStr](#) (System::String^ varName, System::Int32 varScope, System::String^ value)
- System::Boolean [E_ReadPos](#) (System::String^ varName, System::Int32 varScope, [E_ROB_POS_CLI](#)^% valueBack)
- System::Boolean [E_WritePos](#) (System::String^ varName, System::Int32 varScope, [E_ROB_POS_CLI](#)^ value)
- System::Boolean [E_SetMultiDO](#) (System::Collections::Generic::List< System::Int32 >^ IoValues)
- [E_ROB_JOBID_CLI](#) ^ [E_GetCurMovCmdJobId](#) ()
- System::Int32 [E_GetCurMovCmdSts](#) ()
- System::Int32 [E_GetMovCmdSts](#) ([E_ROB_JOBID_CLI](#)^ jobId)
- System::Boolean [E_SetPosArrayValue](#) ([E_VarType_CLI](#) varType, System::String^ varName, [E_ScopeType_CLI](#) varScope, System::Int32 startIndex, System::Int32 nums, cli::array< [E_ROB_POS_CLI](#)^>^ dataIn)
- System::Boolean [E_GetPosArrayValue](#) ([E_VarType_CLI](#) varType, System::String^ varName, [E_ScopeType_CLI](#) varScope, System::Int32 startIndex, System::Int32 nums, cli::array< [E_ROB_POS_CLI](#)^>^% dataOut)
- System::Boolean [E_GetAxeTorqueValue](#) (System::Collections::Generic::List< System::Double >^% data↵ Out)

- `E_ROB_JOBID_CLI ^ E_MovJ2_Trajectory` (cli::array< `E_ROB_POS_CLI`>^ dstPointArray, System::Int32 posNums, [System::Runtime::InteropServices::Optional] `E_ROB_SPEED_CLI`^ velo, [System::Runtime::InteropServices::Optional] `E_ROB_ZONE_CLI`^ zone, [System::Runtime::InteropServices::Optional] System::Int32 payloadId, [System::Runtime::InteropServices::Optional] System::Int32 toolId, [System::Runtime::InteropServices::Optional] System::Int32 coordId, [System::Runtime::InteropServices::Optional] `E_ROB_APIBUFF_CLI`^ addDo, [System::Runtime::InteropServices::Optional] System::Double valfine, [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish)
- `E_ROB_JOBID_CLI ^ E_MovL2_Trajectory` (cli::array< `E_ROB_POS_CLI`>^ dstPointArray, int posNums, [System::Runtime::InteropServices::Optional] `E_ROB_SPEED_CLI`^ velo, [System::Runtime::InteropServices::Optional] `E_ROB_ZONE_CLI`^ zone, [System::Runtime::InteropServices::Optional] System::Int32 payloadId, [System::Runtime::InteropServices::Optional] System::Int32 toolId, [System::Runtime::InteropServices::Optional] System::Int32 coordId, [System::Runtime::InteropServices::Optional] System::Int32 overrideType, [System::Runtime::InteropServices::Optional] `E_ROB_APIBUFF_CLI`^ addDo, [System::Runtime::InteropServices::Optional] System::Double valfine, [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish)
- System::Int32 `E.GetTrajectoryCmdSts` (`E_ROB_JOBID_CLI`^ posArrayJobID)
- System::Boolean `E.GetTrajectoryPointsJobId` (`E_ROB_JOBID_CLI`^ posArrayJobId, cli::array< `E_ROB_JOBID_CLI`>^% posAllJobIDBuff, int buffNums)
- System::Boolean `E.CposToAPos` (System::Collections::Generic::List< System::Double >^ Cpos, System::Collections::Generic::List< System::Double >^% Apos)
- System::Boolean `E.AposToCPos` (System::Collections::Generic::List< System::Double >^ Apos, System::Collections::Generic::List< System::Double >^% Cpos)
- System::Int32 `E.GetCurTcpVelo` ()

5.11.1 详细描述

在文件 `EstunApiForPublic.h` 第 8 行定义.

5.11.2 构造及析构函数说明

5.11.2.1 EstunApi_CLI()

```
EstunApiForPublic::EstunApi_CLI::EstunApi_CLI (
    [System::Runtime::InteropServices::Optional] System::String^ configPath )
```

类的构造和析构接口

参数

<code>configDir</code>	配置文件以及错误信息文件所在文件夹，默认为当前目录下的Config文件夹
------------------------	--------------------------------------

5.11.2.2 ~EstunApi_CLI()

```
EstunApiForPublic::EstunApi_CLI::~~EstunApi_CLI ( )
```


5.11.3 成员函数说明

5.11.3.1 ADDDO_IO_2_E_ROB_APIBUFF()

```
E_ROB_APIBUFF_CLI ^ EstunApiForPublic::EstunApi_CLI::ADDDO_IO_2_E_ROB_APIBUFF (
    E_ROB_ADDDO_IO_CLI^ from )
```

将E_ROB_ADDDO_IO类型转换为E_ROB_APIBUFF.

参数

from	ADD.DO.IO参数
------	-------------

返回

ADD.DO结构

5.11.3.2 connectToServer()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::connectToServer (
    System::String^ ipAddr,
    System::Boolean isAutoReConnect )
```

连接服务器.

参数

ipAddr	服务IP地址, IPV4格式
isAutoReConnect	是否自动重连, 如果为True, 客户端在异常掉线时会自动重连, 否则不会

返回

成功返回true, 失败返回false

5.11.3.3 disConnect()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::disConnect ( )
```

断开与服务的连接.

返回

成功返回true, 失败返回false

5.11.3.4 E.AcquirePermit()

```
E.ROB.PERMIT.CLI ^ EstunApiForPublic::EstunApi_CLI::E.AcquirePermit ( )
```

申请机器人运动操作权限.

返回
 权限信息

5.11.3.5 E.AposToCPos()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.AposToCPos (
    System::Collections::Generic::List< System::Double >^ Apos,
    System::Collections::Generic::List< System::Double >^% Cpos )
```

关节坐标转笛卡尔坐标.

参数

<i>Apos</i>	关节坐标 a1,a2,a3,a4,a5,a6
<i>Cpos</i>	笛卡尔坐标 mode,cf1,cf2,cf3,cf4,cf5,cf6,x,y,z,a,b,c

返回
 转换成功返回true

5.11.3.6 E.CheckClearError()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.CheckClearError ( )
```

清除所有错误报警后获取是否还有错误.

返回
 没有错误返回true，有错误返回false

5.11.3.7 E.ClearError()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.ClearError ( )
```

清除错误报警.

返回
 成功返回true，失败返回false

5.11.3.8 E.CposToAPos()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.CposToAPos (
    System::Collections::Generic::List< System::Double >^ Cpos,
    System::Collections::Generic::List< System::Double >^% Apos )
```

笛卡尔坐标转关节坐标.

参数

<i>Cpos</i>	笛卡尔坐标 mode,cf1,cf2,cf3,cf4,cf5,cf6,x,y,z,a,b,c
<i>Apos</i>	关节坐标 a1,a2,a3,a4,a5,a6

返回

转换成功返回true

5.11.3.9 E.CreatePayLoad()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.CreatePayLoad (
    System::Int32 payloadID,
    E_ROB_PAYLOAD_CLI^ payloadData )
```

创建负载.

参数

<i>payloadID</i>	新的负载号
<i>payloadData</i>	负载参数

返回

成功返回true，失败返回false

5.11.3.10 E.CreateTool()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.CreateTool (
    System::Int32 toolID,
    E_ROB_COORD_CLI^ toolCoord,
    E_ROB_PAYLOAD_CLI^ payloadvalue )
```

创建工具参数.

参数

<i>toolID</i>	新的工具号
<i>toolCoord</i>	工具坐标参数
<i>payloadvalue</i>	工具负载参数

返回

成功返回true，失败返回false

5.11.3.11 E.CreateUserCoord()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.CreateUserCoord (
    System::Int32 usrID,
    E_ROB_COORD_CLI^ usrData )
```

创建用户坐标.

参数

<i>usrID</i>	新的坐标号
<i>usrData</i>	坐标参数

返回

成功返回true，失败返回false

5.11.3.12 E.CurrentPermit()

```
E_ROB_PERMIT_CLI ^ EstunApiForPublic::EstunApi_CLI::E.CurrentPermit ( )
```

查询当前机器人运动操作权限.

返回

权限信息

5.11.3.13 E.GetAPIStatus()

```
System::Int32 EstunApiForPublic::EstunApi_CLI::E.GetAPIStatus ( )
```

获取当前模式的工作状态.

返回

-1表示失败, 0表示正常, 1表示机器人错误, 2表示机器人处于停止状态, 需要调用start接口启动

5.11.3.14 E.GetAxeTorqueValue()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetAxeTorqueValue (
    System::Collections::Generic::List< System::Double >^% dataOut )
```

获取力矩的值.

参数

<i>dataOut</i>	从力矩值的存储位置, 需要预先申请内存, 16个double的内存。
----------------	------------------------------------

返回

成功返回true, 失败返回false

5.11.3.15 E.GetCurJPos()

```
E.ROB_POS_CLI ^ EstunApiForPublic::EstunApi_CLI::E.GetCurJPos ( )
```

获取当前位置信息, 关节坐标系.

返回

位置信息, APos

5.11.3.16 E.GetCurMovCmdJobId()

```
E.ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E.GetCurMovCmdJobId ( )
```

获取最近一次下发成功的运动指令的信息.

返回

运动指令信息

5.11.3.17 E.GetCurMovCmdSts()

```
System::Int32 EstunApiForPublic::EstunApi_CLI::E.GetCurMovCmdSts ( )
```

获取最近一次下发成功的运动指令的执行状态.

返回

-1-未找到, 0-运动未完成, 1-运动完成

5.11.3.18 E.GetCurPayLoadID()

```
System::Int32 EstunApiForPublic::EstunApi_CLI::E.GetCurPayLoadID ( )
```

获取当前负载号.

返回

负载号

5.11.3.19 E.GetCurProgramName()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetCurProgramName (
    E_ROB_PROGNAME_CLI ^ % value )
```

获取当前加载的程序名.

参数

<i>value</i>	工程程序名, 引用类型
--------------	-------------

返回

成功返回true, 失败返回false

5.11.3.20 E.GetCurProgramRunStatus()

```
E.RunStatusType_CLI EstunApiForPublic::EstunApi_CLI::E.GetCurProgramRunStatus ( )
```

获取当前程序运行状态.

返回

程序运行状态

5.11.3.21 E.GetCurRunMode()

`E.RunModeType_CLI` EstunApiForPublic::EstunApi_CLI::E.GetCurRunMode ()

获取运动模式.

返回

运动模式

5.11.3.22 E.GetCurTcpVelo()

`System::Int32` EstunApiForPublic::EstunApi_CLI::E.GetCurTcpVelo ()

获得机器人当前线速度

返回

返回机器人线速度

5.11.3.23 E.GetCurToolID()

`System::Int32` EstunApiForPublic::EstunApi_CLI::E.GetCurToolID ()

获取当前工具号.

返回

工具号

5.11.3.24 E.GetCurUserCoordID()

`System::Int32` EstunApiForPublic::EstunApi_CLI::E.GetCurUserCoordID (
 `E.UserCoordType_CLI` *coordtype*)

获取当前用户坐标号.

参数

<code>coordtype</code>	用户坐标类型, 引用
------------------------	------------

返回

用户坐标号

5.11.3.25 E.GetCurWPos()

```
E_ROB_POS_CLI ^ EstunApiForPublic::EstunApi_CLI::E.GetCurWPos ( )
```

获取当前位置信息，世界坐标系.

返回

位置信息，CPos

5.11.3.26 E.GetDefaultPayload()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetDefaultPayload (
    System::Int32% payloadID,
    System::String^% name,
    E_ROB_PAYLOAD_CLI^% payload )
```

获取默认加载的负载参数.

参数

<i>payloadID</i>	负载id，引用
<i>name</i>	负载名称，引用
<i>payload</i>	负载参数，引用

返回

成功返回true，失败返回false

5.11.3.27 E.GetErrorId()

```
System::Int32 EstunApiForPublic::EstunApi_CLI::E.GetErrorId ( )
```

获取机器人错误号.

返回

错误号

5.11.3.28 E.GetErrorString() [1/2]

```
System::String ^ EstunApiForPublic::EstunApi_CLI::E.GetErrorString (
    System::Int32 errorId )
```

获取机器人错误内容.

参数

<i>errorId</i>	错误号
----------------	-----

返回

错误详情

5.11.3.29 E.GetErrorString() [2/2]

```
System::String ^ EstunApiForPublic::EstunApi_CLI::E.GetErrorString (
    System::Int32 errorId,
    System::String^ language )
```

获取机器人错误内容.

参数

<i>errorId</i>	错误号
<i>language</i>	语言，目前支持ch-中文，en-英文

返回

错误详情

5.11.3.30 E.GetFine()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetFine (
    System::Double % value )
```

获取默认到位精度值.

参数

<i>value</i>	到位精度值，引用类型
--------------	------------

返回

成功返回true，失败返回false

5.11.3.31 E.GetGlobalSpeed()

```
System::Int32 EstunApiForPublic::EstunApi_CLI::E.GetGlobalSpeed ( )
```

获取全局速度.

返回

速度值

5.11.3.32 E.GetIOValue()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetIOValue (
    System::Int32 IoIndex,
    E.IOType_CLI IoType,
    System::Int32 IoNum,
    System::Collections::Generic::List< System::Double >^% dataOut )
```

获取IO的值.

参数

<i>IoIndex</i>	开始的IO下标
<i>IoType</i>	IO类型
<i>IoNum</i>	查询IO数量
<i>dataOut</i>	从【IoIndex】开始的【IoNum】个的IO的值数组，引用类型

返回

成功返回true，失败返回false

5.11.3.33 E.GetJogCoord()

```
E.CoordType_CLI EstunApiForPublic::EstunApi_CLI::E.GetJogCoord ( )
```

获取点动坐标系类型.

返回

坐标系类型

5.11.3.34 E.GetJogEnable()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetJogEnable ( )
```

获取当前模式下job是否可用.

返回

可用返回true, 不可用返回false

5.11.3.35 E.GetMovCmdSts()

```
System::Int32 EstunApiForPublic::EstunApi_CLI::E.GetMovCmdSts (
    E_ROB_JOBID_CLI^ jobID )
```

获取指定运动指令执行状态.

参数

jobID	
-------	--

返回

-1-未找到, 0-运动未完成, 1-运动完成,2-暂停, 3-结束/错误, 4-(预留类型)

5.11.3.36 E.GetPayLoad()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetPayLoad (
    System::Int32 payloadID,
    E_ROB_PAYLOAD_CLI^% payload )
```

获取负载信息.

参数

payloadID	负载变量id号
payload	负载参数, 引用

返回

成功返回true, 失败返回false

5.11.3.37 E.GetPayLoadName()

```
System::String ^ EstunApiForPublic::EstunApi_CLI::E.GetPayLoadName (
    System::Int32 payloadID )
```

获取负载变量名称.

参数

<i>payloadID</i>	负载变量id
------------------	--------

返回

名称, 为空则表示获取失败

5.11.3.38 E.GetPayLoadsID()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetPayLoadsID (
    System::Int32% num,
    System::Collections::Generic::List< System::Int32 >^% idArray )
```

获取所有负载ID.

参数

<i>num</i>	获取到的个数
<i>idArray</i>	ID数组指针, 由接口初始化

返回

成功获取返回true

5.11.3.39 E.GetPosArrayValue()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetPosArrayValue (
    E.VarType_CLI varType,
    System::String^ varName,
    E.ScopeType_CLI varScope,
    System::Int32 startIndex,
    System::Int32 nums,
    cli::array< E.ROB.POS_CLI>^% dataOut )
```

获取点位数组.

参数

<i>varType</i>	变量类型 只可以是CposArray或AposArray,其他类型无效
<i>varName</i>	变量名
<i>varScope</i>	变量域
<i>startIndex</i>	指定存储的起始下标
<i>nums</i>	指定存储的个数
<i>dataOut</i>	返回的点位数组

返回

成功返回true，失败返回false

5.11.3.40 E.GetRobotInfo()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetRobotInfo (
    System::String^% value )
```

获取机器人信息.

参数

<i>value</i>	机器人信息
--------------	-------

返回

成功返回true，失败返回false

5.11.3.41 E.GetRobotRunStatus()

```
System::Int32 EstunApiForPublic::EstunApi_CLI::E.GetRobotRunStatus ( )
```

获取机器人运动状态.

返回

运动返回1，不运动返回0

5.11.3.42 E.GetRobotStatus()

```
System::Int32 EstunApiForPublic::EstunApi_CLI::E.GetRobotStatus ( )
```

获取机器人状态.

参数

<i>value</i>	机器人状态，12345表示在线，其余为不在线
--------------	------------------------

5.11.3.43 E.GetRobotType()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetRobotType (
    System::String^% value )
```

获取机器人类型.

参数

<i>value</i>	机器人型号
--------------	-------

返回

成功返回true，失败返回false

5.11.3.44 E.GetServoOn()

```
E.ServoStatusType_CLI EstunApiForPublic::EstunApi_CLI::E.GetServoOn ( )
```

获取机器人伺服是否使能.

返回

使能标志

5.11.3.45 E.GetSysMode()

```
E.SysModeType_CLI EstunApiForPublic::EstunApi_CLI::E.GetSysMode ( )
```

获取系统模式.

返回

系统模式

5.11.3.46 E.GetTool()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetTool (
    System::Int32 toolID,
    E_ROB_COORD_CLI^% toolCoord,
    E_ROB_PAYLOAD_CLI^% toolPayload )
```

获取工具信息.

参数

<i>toolID</i>	工具号
<i>toolCoord</i>	工具参数，引用
<i>toolPayload</i>	负载参数，引用

返回

成功返回true，失败返回false

5.11.3.47 E.GetToolName()

```
System::String ^ EstunApiForPublic::EstunApi_CLI::E.GetToolName (
    System::Int32 toolID )
```

获取工具坐标变量名称.

参数

<i>toolID</i>	工具id
---------------	------

返回

名称，为空则表示获取失败

5.11.3.48 E.GetToolsID()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetToolsID (
    System::Int32% num,
    System::Collections::Generic::List< System::Int32 >^% idArray )
```

获取所有工具坐标系ID.

参数

<i>num</i>	获取到的个数
<i>idArray</i>	ID数组指针，由接口初始化

返回

成功获取返回true

5.11.3.49 E.GetTrajectoryCmdSts()

```
System::Int32 EstunApiForPublic::EstunApi_CLI::E.GetTrajectoryCmdSts (
    E_ROB_JOBID_CLI^ posArrayJobID )
```

获取指定运动数组指令执行状态.

参数

<i>posArrayJobID</i>	指定的轨迹id信息
----------------------	-----------

返回

-2-错误 -1-未找到, 0-运动未完成, 1-运动完成,2-暂停, 3-结束/运动错误, 4-(预留类型)

5.11.3.50 E.GetTrajectoryPointsJobId()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetTrajectoryPointsJobId (
    E_ROB_JOBID_CLI^ posArrayJobID,
    cli::array< E_ROB_JOBID_CLI^>^% posAllJobIDBuff,
    int buffNums )
```

获取指定运动数组指令执行状态.

参数

<i>posArrayJobID</i>	指定的轨迹id信息
<i>posAllJobIDBuff</i>	接收获取到的运动信息的数组空间
<i>buffNums</i>	数组容量, 最大50个

返回

true-成功,false-失败

5.11.3.51 E.GetUserCoord()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetUserCoord (
    System::Int32 coordID,
    E_UserCoordType_CLI coordtype,
    E_ROB_COORD_CLI^% coord )
```

获取用户坐标信息.

参数

<i>coordID</i>	用户坐标号
<i>coordtype</i>	用户坐标类型
<i>coord</i>	用户坐标参数，引用

返回

成功返回true，失败返回false

5.11.3.52 E.GetUserCoordName()

```
System::String ^ EstunApiForPublic::EstunApi_CLI::E.GetUserCoordName (
    System::Int32 coordID,
    E.UserCoordType_CLI coordtype )
```

获取用户坐标变量名称.

参数

<i>coordID</i>	用户坐标id
<i>coordtype</i>	用户坐标类型

返回

名称，为空则表示获取失败

5.11.3.53 E.GetUserCoordsID()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetUserCoordsID (
    System::Int32% num,
    System::Collections::Generic::List< System::Int32 >^% idArray )
```

获取所有用户坐标系ID. (ID大于10000为外部坐标系)

参数

<i>num</i>	获取到的个数
<i>idArray</i>	ID数组指针，由接口初始化

返回

成功获取返回true

5.11.3.54 E.GetVarValue()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.GetVarValue (
    E.VarType_CLI varType,
    System::String^ varName,
    E.ScopeType_CLI varScope,
    System::String^% valueBack )
```

获取变量.

参数

<i>varType</i>	变量类型
<i>varName</i>	变量名
<i>varScope</i>	变量域
<i>valueBack</i>	变量值，字符串类型，引用类型，需要转换为其他类型

返回

成功返回true，失败返回false

5.11.3.55 E.JogMicroMotion()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.JogMicroMotion (
    System::Int32 axisId,
    E.DistentionType_CLI distention,
    System::Double dist,
    System::Boolean mov )
```

寸动.

参数

<i>axisId</i>	值[0,5]，对应关节坐标系下为轴1-6，世界坐标系下为xyzabc
<i>distention</i>	运动方向
<i>dist</i>	寸动距离，单位毫米
<i>mov</i>	为true运动，为false停止

返回

成功返回true，失败返回false

5.11.3.56 E_JogMotion()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_JogMotion (
    System::Int32 axisId,
    E_DistentionType_CLI distention,
    System::Boolean mov )
```

点动.

参数

axisId	值[0,5]，对应关节坐标系下为轴1-6，世界坐标系下为xyzabc
distention	运动方向
mov	为true运动，为false停止

返回

成功返回true，失败返回false

5.11.3.57 E_LoadPayLoad()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_LoadPayLoad (
    System::Int32 payloadID )
```

加载负载.

参数

payloadID	负载号
-----------	-----

返回

成功返回true，失败返回false

5.11.3.58 E_LoadProgame()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_LoadProgame (
    System::String^ prjname,
    System::String^ programname )
```

加载程序.

参数

<i>prjname</i>	工程名
<i>programname</i>	程序名

返回

成功返回true, 失败返回false

5.11.3.59 E.LoadProject()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.LoadProject (
    System::String^ prjname )
```

只加载工程.

参数

<i>prjname</i>	工程名
----------------	-----

返回

成功返回true, 失败返回false

5.11.3.60 E.LoadTool()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.LoadTool (
    System::Int32 toolID )
```

加载工具.

参数

<i>toolID</i>	工具号
---------------	-----

返回

成功返回true, 失败返回false

5.11.3.61 E.LoadUserCoord()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.LoadUserCoord (
    System::Int32 usrID )
```

加载用户坐标.

参数

<i>usrID</i>	用户坐标号
--------------	-------

返回

成功返回true，失败返回false

5.11.3.62 E.ModifyPayLoad()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.ModifyPayLoad (
    System::Int32 payloadID,
    E_ROB_PAYLOAD_CLI^ payloadData )
```

修改负载.

参数

<i>payloadID</i>	要修改的负载号
<i>payloadData</i>	负载参数

返回

成功返回true，失败返回false

5.11.3.63 E.ModifyTool()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.ModifyTool (
    System::Int32 toolID,
    E_ROB_COORD_CLI^ toolCoord,
    E_ROB_PAYLOAD_CLI^ payloadvalue )
```

修改工具参数.

参数

<i>toolID</i>	要修改的工具号
<i>toolCoord</i>	工具坐标参数
<i>payloadvalue</i>	工具负载参数

返回

成功返回true，失败返回false

5.11.3.64 E.ModifyUserCoord()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.ModifyUserCoord (
    System::Int32  usrID,
    E_ROB_COORD_CLI^ usrData )
```

修改用户坐标.

参数

usrID	要修改的用户坐标号
usrData	坐标参数

返回

成功返回true，失败返回false

5.11.3.65 E.MotionContinue()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.MotionContinue ( )
```

恢复机器人运动，继续执行运动队列.

返回

成功返回true，失败返回false

5.11.3.66 E.MotionPause()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.MotionPause ( )
```

暂停机器人运动，不会清空运动队列.

返回

成功返回true，失败返回false

5.11.3.67 E.MotionStart()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.MotionStart ( )
```

唤醒机器人运动，从stop模式恢复.

返回

成功返回true，失败返回false

5.11.3.68 E.MotionStop()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.MotionStop ( )
```

停止机器人运动，同时清空运动队列.

返回

成功返回true，失败返回false

5.11.3.69 E.MovArch()

```
E.ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E.MovArch (
    System::String^ Pname,
    [System::Runtime::InteropServices::Optional] System::String^ velName,
    [System::Runtime::InteropServices::Optional] System::String^ blendType,
    [System::Runtime::InteropServices::Optional] System::String^ zname,
    [System::Runtime::InteropServices::Optional] System::String^ tname,
    [System::Runtime::InteropServices::Optional] System::String^ coorname,
    [System::Runtime::InteropServices::Optional] System::String^ payloadName,
    [System::Runtime::InteropServices::Optional] System::String^ cornerName,
    [System::Runtime::InteropServices::Optional] System::Double height,
    [System::Runtime::InteropServices::Optional] E.ROB_APIBUFF_CLI^ addDo,
    [System::Runtime::InteropServices::Optional] System::Double valfine,
    [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )
```

执行MovArch指令.

参数

<i>Pname</i>	目标位置点变量名， 例: "t.g.p1"
<i>velName</i>	运动速度变量名， 例: "t.g.SPEED0"
<i>blendType</i>	过渡类型， "", "Rel", "Abs", 三选一
<i>zname</i>	过渡参数变量名， 例: "t.g.ZONE0"
<i>tname</i>	工具变量名， 例: "t.g.TOOL1"
<i>coorname</i>	坐标系变量名， 例: "t.g.USER0"
<i>payloadName</i>	负载变量名， 例: "t.g. PAYLOAD0"
<i>cornerName</i>	拱形运动转角变量名， 例: "t.g.ZONE1"
<i>height</i>	抬升高度， 单位毫米
<i>addDo</i>	附加指令， 内容由其他结构体填充， 需要调用AddDoTypeConvert的相关接口完成转

返回

指令标识信息，id小于0为失败

5.11.3.70 E.MovArch2()

```
E_ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E_MovArch2 (
    E_ROB_POS_CLI^ dstPoint,
    [System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
    [System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
    [System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ connerZone,
    [System::Runtime::InteropServices::Optional] System::Int32 payloadId,
    [System::Runtime::InteropServices::Optional] System::Int32 toolId,
    [System::Runtime::InteropServices::Optional] System::Int32 coordId,
    [System::Runtime::InteropServices::Optional] System::Double height,
    [System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
    [System::Runtime::InteropServices::Optional] System::Double valfine,
    [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )
```

执行MovArch2指令.

参数

<i>dstPoint</i>	目标位置点
<i>velo</i>	运行速度
<i>zone</i>	过度参数
<i>connerZone</i>	转角参数
<i>payloadId</i>	负载号
<i>toolId</i>	工具号
<i>coordId</i>	用户坐标号
<i>height</i>	抬升高度，单位毫米
<i>addDo</i>	附加指令，内容由其他结构体填充，需要调用AddDoTypeConvert的相关接口完成转换，目前仅支持E_ROB_ADDDO_IO类型
<i>valfine</i>	到位精度值
<i>IsWaitFinish</i>	是否等运动执行结束，为true则接口在运动执行到位后再返回，为false则立即返回

返回

指令标识信息，id小于0为失败

5.11.3.71 E.MovC()

```
E_ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E_MovC (
    System::String^ PAuxname,
    System::String^ PDstname,
```



```

[System::Runtime::InteropServices::Optional] System::String^ velName,
[System::Runtime::InteropServices::Optional] System::String^ blendType,
[System::Runtime::InteropServices::Optional] System::String^ zname,
[System::Runtime::InteropServices::Optional] System::String^ tname,
[System::Runtime::InteropServices::Optional] System::String^ coordname,
[System::Runtime::InteropServices::Optional] System::String^ payloadName,
[System::Runtime::InteropServices::Optional] System::String^ overrideType,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )

```

执行MovC指令.

参数

<i>PAuxname</i>	辅助位置点变量名, 例: "t.g.p0"
<i>PDstname</i>	目标位置点变量名, 例: "t.g.p1"
<i>velName</i>	运动速度变量名, 例: "t.g.SPEED0"
<i>blendType</i>	过渡类型, "", "Rel", "Abs", 三选一
<i>zname</i>	过渡参数变量名, 例: "t.g.ZONE0"
<i>tname</i>	工具变量名, 例: "t.g.TOOL1"
<i>coordname</i>	坐标系变量名, 例: "t.g.USER0"
<i>payloadName</i>	负载变量名, 例: "t.g. PAYLOAD0"
<i>overrideType</i>	速率类型, "", "GOFF", "GON", 三选一
<i>addDo</i>	附加指令, 内容由其他结构体填充, 需要调用AddDoTypeConvert的相关接口完成转换, 目前仅支持E_ROB.ADDDDO.IO类型
<i>valfine</i>	到位精度值
<i>IsWaitFinish</i>	是否等运动执行结束, 为true则接口在运动执行到位后再返回, 为false则立即返回

返回

指令标识信息, id小于0为失败

5.11.3.72 E_MovC2()

```

E_ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E_MovC2 (
    E_ROB_POS_CLI^ auxP,
    E_ROB_POS_CLI^ dstPoint,
[System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
[System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
[System::Runtime::InteropServices::Optional] System::Int32 payloadId,
[System::Runtime::InteropServices::Optional] System::Int32 toolId,
[System::Runtime::InteropServices::Optional] System::Int32 coordId,
[System::Runtime::InteropServices::Optional] System::Int32 overrideType,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )

```

执行MovC2指令.

参数

<i>auxP</i>	辅助位置点
<i>dstPoint</i>	目标位置点
<i>velo</i>	运行速度
<i>zone</i>	过度参数
<i>payloadId</i>	负载号
<i>toolId</i>	工具号
<i>coordId</i>	用户坐标号
<i>overrideType</i>	速率类型, 0:GOVROFF;1:GOVRON
<i>addDo</i>	附加指令，内容由其他结构体填充，需要调用AddDoTypeConvert的相关接口完成转换，目前仅支持E_ROB_ADDDO_IO类型
<i>valfine</i>	到位精度值
<i>IsWaitFinish</i>	是否等运动执行结束，为true则接口在运动执行到位后再返回，为false则立即返回

返回

指令标识信息，id小于0为失败

5.11.3.73 E.MovJ()

```
E_ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E.MovJ (
    System::String^ Pname,
    [System::Runtime::InteropServices::Optional] System::String^ velName,
    [System::Runtime::InteropServices::Optional] System::String^ blendType,
    [System::Runtime::InteropServices::Optional] System::String^ zname,
    [System::Runtime::InteropServices::Optional] System::String^ tname,
    [System::Runtime::InteropServices::Optional] System::String^ coordname,
    [System::Runtime::InteropServices::Optional] System::String^ payloadName,
    [System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
    [System::Runtime::InteropServices::Optional] System::Double valfine,
    [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )
```

执行MovJ指令.

参数

<i>Pname</i>	目标位置点变量名，例: "t.g.p1"
<i>velName</i>	运动速度变量名，例: "t.g.SPEED0"
<i>blendType</i>	过渡类型, "", "Rel", "Abs", 三选一
<i>zname</i>	过渡参数变量名，例: "t.g.ZONE0"
<i>tname</i>	工具变量名，例: "t.g.TOOL1"
<i>coordname</i>	坐标系变量名，例: "t.g.USER0"
<i>payloadName</i>	负载变量名，例: "t.g. PAYLOAD0"
<i>addDo</i>	附加指令，内容由其他结构体填充，需要调用AddDoTypeConvert的相关接口完成转换，目前仅支持E_ROB_ADDDO_IO类型
<i>valfine</i>	到位精度值
<i>IsWaitFinish</i>	是否等运动执行结束，为true则接口在运动执行到位后再返回，为false则立即返回

返回

指令标识信息，id小于0为失败

5.11.3.74 E.MovJ2()

```
E_ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E_MovJ2 (
    E_ROB_POS_CLI^ dstPoint,
    [System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
    [System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
    [System::Runtime::InteropServices::Optional] System::Int32 payloadId,
    [System::Runtime::InteropServices::Optional] System::Int32 toolId,
    [System::Runtime::InteropServices::Optional] System::Int32 coordId,
    [System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
    [System::Runtime::InteropServices::Optional] System::Double valfine,
    [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )
```

执行MovJ2指令.

参数

<i>dstPoint</i>	目标位置点
<i>velo</i>	运行速度
<i>zone</i>	过度参数
<i>payloadId</i>	负载号
<i>toolId</i>	工具号
<i>coordId</i>	用户坐标号
<i>addDo</i>	附加指令，内容由其他结构体填充，需要调用AddDoTypeConvert的相关接口完成转换，目前仅支持E_ROB_ADDDO_IO类型
<i>valfine</i>	到位精度值
<i>IsWaitFinish</i>	是否等运动执行结束，为true则接口在运动执行到位后再返回，为false则立即返回

返回

指令标识信息，id小于0为失败

5.11.3.75 E.MovJ2_Trajectory()

```
E_ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E_MovJ2_Trajectory (
    cli::array< E_ROB_POS_CLI^>^ dstPointArray,
    System::Int32 posNums,
    [System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
    [System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
    [System::Runtime::InteropServices::Optional] System::Int32 payloadId,
    [System::Runtime::InteropServices::Optional] System::Int32 toolId,
```

```
[System::Runtime::InteropServices::Optional] System::Int32 coordId,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )
```

执行MovJ2数组指令.

参数

<i>dstPointArray</i>	目标位置点,轨迹点数组
<i>posNums</i>	轨迹点个数, 最大支持50个
<i>velo</i>	运行速度
<i>zone</i>	过度参数
<i>payloadId</i>	负载号
<i>toolId</i>	工具号
<i>coordId</i>	用户坐标号
<i>addDo</i>	附加指令, 内容由其他结构体填充, 需要调用AddDoTypeConvert的相关接口完成转换, 目前仅支持E_ROB_ADDDO_IO类型
<i>valfine</i>	到位精度值
<i>IsWaitFinish</i>	是否等运动执行结束, 为true则接口在运动执行到位后再返回, 为false则立即返回

返回

指令标识信息, id小于0为失败

5.11.3.76 E.MovJArch2()

```
E_ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E_MovJArch2 (
    E_ROB_POS_CLI^ dstPoint,
[System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
[System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
[System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ connerZone,
[System::Runtime::InteropServices::Optional] System::Int32 payloadId,
[System::Runtime::InteropServices::Optional] System::Int32 toolId,
[System::Runtime::InteropServices::Optional] System::Int32 coordId,
[System::Runtime::InteropServices::Optional] System::Double Upheight,
[System::Runtime::InteropServices::Optional] System::Double Downheight,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )
```

执行E_MovJArch2指令.

参数

<i>dstPoint</i>	目标位置点
<i>velo</i>	运行速度
<i>zone</i>	过度参数

参数

<i>connerZone</i>	转角参数
<i>payloadId</i>	负载号
<i>toolId</i>	工具号
<i>coordId</i>	用户坐标号
<i>Upheight</i>	抬升高度，单位毫米
<i>Downheight</i>	抬升高度，单位毫米
<i>addDo</i>	附加指令，内容由其他结构体填充，需要调用AddDoTypeConvert的相关接口完成转换，目前仅支持E.ROB.ADDDO.IO类型
<i>valfine</i>	到位精度值
<i>IsWaitFinish</i>	是否等运动执结束，为true则接口在运动执行到位后再返回，为false则立即返回

返回

指令标识信息，id小于0为失败

5.11.3.77 E.MovL()

```

E.ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::EMovL (
    System::String^ Pname,
    [System::Runtime::InteropServices::Optional] System::String^ velName,
    [System::Runtime::InteropServices::Optional] System::String^ blendType,
    [System::Runtime::InteropServices::Optional] System::String^ zname,
    [System::Runtime::InteropServices::Optional] System::String^ tname,
    [System::Runtime::InteropServices::Optional] System::String^ coordname,
    [System::Runtime::InteropServices::Optional] System::String^ payloadName,
    [System::Runtime::InteropServices::Optional] System::String^ overrideType,
    [System::Runtime::InteropServices::Optional] E.ROB_APIBUFF_CLI^ addDo,
    [System::Runtime::InteropServices::Optional] System::Double valfine,
    [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )

```

执行MovL指令.

参数

<i>Pname</i>	目标位置点变量名，例: "t.g.p1"
<i>velName</i>	运动速度变量名，例: "t.g.SPEED0"
<i>blendType</i>	过渡类型，"","Rel","Abs"，三选一
<i>zname</i>	过渡参数变量名，例: "t.g.ZONE0"
<i>tname</i>	工具变量名，例: "t.g.TOOL1"
<i>coordname</i>	坐标系变量名，例: "t.g.USER0"
<i>payloadName</i>	负载变量名，例: "t.g. PAYLOAD0"
<i>overrideType</i>	速率类型，"","GOFF","GON"，三选一
<i>addDo</i>	附加指令，内容由其他结构体填充，需要调用AddDoTypeConvert的相关接口完成转换，目前仅支持E.ROB.ADDDO.IO类型
<i>valfine</i>	到位精度值
<i>IsWaitFinish</i>	是否等运动执行结束，为true则接口在运动执行到位后再返回，为false则立即返回

返回

指令标识信息，id小于0为失败

5.11.3.78 E.MovL2()

```
E_ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E_MovL2 (
    E_ROB_POS_CLI^ dstPoint,
    [System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
    [System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
    [System::Runtime::InteropServices::Optional] System::Int32 payloadId,
    [System::Runtime::InteropServices::Optional] System::Int32 toolId,
    [System::Runtime::InteropServices::Optional] System::Int32 coordId,
    [System::Runtime::InteropServices::Optional] System::Int32 overrideType,
    [System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
    [System::Runtime::InteropServices::Optional] System::Double valfine,
    [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )
```

执行MovL2指令.

参数

<i>dstPoint</i>	目标位置点
<i>velo</i>	运行速度
<i>zone</i>	过度参数
<i>payloadId</i>	负载号
<i>toolId</i>	工具号
<i>coordId</i>	用户坐标号
<i>overrideType</i>	速率类型, 0:GOVROFF;1:GOVRON
<i>addDo</i>	附加指令，内容由其他结构体填充，需要调用AddDoTypeConvert的相关接口完成转换，目前仅支持E_ROB_ADDDO_IO类型
<i>valfine</i>	到位精度值
<i>IsWaitFinish</i>	是否等运动执行结束，为true则接口在运动执行到位后再返回，为false则立即返回

返回

指令标识信息，id小于0为失败

5.11.3.79 E.MovL2.Trajectory()

```
E_ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E_MovL2_Trajectory (
    cli::array< E_ROB_POS_CLI^>^ dstPointArray,
    int posNums,
    [System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
    [System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
```

```
[System::Runtime::InteropServices::Optional] System::Int32 payloadId,
[System::Runtime::InteropServices::Optional] System::Int32 toolId,
[System::Runtime::InteropServices::Optional] System::Int32 coordId,
[System::Runtime::InteropServices::Optional] System::Int32 overrideType,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )
```

执行MovL2指令数组.

参数

<i>dstPointArray</i>	目标位置点,轨迹点数组
<i>posNums</i>	轨迹点的个数, 最大支持50个
<i>velo</i>	运行速度
<i>zone</i>	过度参数
<i>payloadId</i>	负载号
<i>toolId</i>	工具号
<i>coordId</i>	用户坐标号
<i>overrideType</i>	速率类型, 0:GOVROFF;1:GOVRON
<i>addDo</i>	附加指令, 内容由其他结构体填充, 需要调用AddDoTypeConvert的相关接口完成转换, 目前仅支持E_ROB_ADDDO_IO类型
<i>valfine</i>	到位精度值
<i>IsWaitFinish</i>	是否等运动执行结束, 为true则接口在运动执行到位后再返回, 为false则立即返回

返回

指令标识信息, id小于0为失败

5.11.3.80 E.MovLArch2()

```
E_ROB_JOBID_CLI ^ EstunApiForPublic::EstunApi_CLI::E_MovLArch2 (
    E_ROB_POS_CLI^ dstPoint,
    [System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
    [System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
    [System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ connerZone,
    [System::Runtime::InteropServices::Optional] System::Int32 payloadId,
    [System::Runtime::InteropServices::Optional] System::Int32 toolId,
    [System::Runtime::InteropServices::Optional] System::Int32 coordId,
    [System::Runtime::InteropServices::Optional] System::Double Upheight,
    [System::Runtime::InteropServices::Optional] System::Double Downheight,
    [System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
    [System::Runtime::InteropServices::Optional] System::Double valfine,
    [System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish )
```

执行E_MovLArch2指令.

参数

<i>dstPoint</i>	目标位置点
<i>velo</i>	运行速度
<i>zone</i>	过度参数
<i>connerZone</i>	转角参数
<i>payloadId</i>	负载号
<i>toolId</i>	工具号
<i>coordId</i>	用户坐标号
<i>Upheight</i>	抬升高度，单位毫米
<i>Downheight</i>	抬升高度，单位毫米
<i>addDo</i>	附加指令，内容由其他结构体填充，需要调用AddDoTypeConvert的相关接口完成转换，目前仅支持E_ROB_ADDDO_IO类型
<i>valfine</i>	到位精度值
<i>IsWaitFinish</i>	是否等运动执结束，为true则接口在运动执行到位后再返回，为false则立即返回

返回

指令标识信息，id小于0为失败

5.11.3.81 E.ProgPause()

System::Boolean EstunApiForPublic::EstunApi_CLI::E.ProgPause ()

暂停程序.

返回

成功返回true，失败返回false

5.11.3.82 E.ProgRun()

System::Boolean EstunApiForPublic::EstunApi_CLI::E.ProgRun ()

运行程序.

返回

成功返回true，失败返回false

5.11.3.83 E.ProgStop()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.ProgStop ( )
```

停止程序.

返回

成功返回true, 失败返回false

5.11.3.84 E.ReadInt()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.ReadInt (
    System::String^ varName,
    System::Int32 varScope,
    System::Int32 % valueBack )
```

读取Int型变量值.

参数

<i>varName</i>	变量名称
<i>varScope</i>	变量域
<i>valueBack</i>	变量值, 引用类型

返回

成功返回true, 失败返回false

5.11.3.85 E.ReadPos()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.ReadPos (
    System::String^ varName,
    System::Int32 varScope,
    E_ROB_POS_CLI^% valueBack )
```

读取位置类型变量值.

参数

<i>varName</i>	变量名称
<i>varScope</i>	变量域
<i>valueBack</i>	变量值, 引用类型

返回

成功返回true，失败返回false

5.11.3.86 E.ReadReal()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.ReadReal (
    System::String^ varName,
    System::Int32 varScope,
    System::Double % valueBack )
```

读取Double型变量值.

参数

<i>varName</i>	变量名称
<i>varScope</i>	变量域
<i>valueBack</i>	变量值，引用类型

返回

成功返回true，失败返回false

5.11.3.87 E.ReadStr()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.ReadStr (
    System::String^ varName,
    System::Int32 varScope,
    System::String^% valueBack )
```

读取String型变量值.

参数

<i>varName</i>	变量名称
<i>varScope</i>	变量域
<i>valueBack</i>	变量值，引用类型

返回

成功返回true，失败返回false

5.11.3.88 E.ReleasePermit()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.ReleasePermit (
    E_ROB_PERMIT_CLI^ permit )
```

释放机器人运动操作权限.

参数

<i>permit</i>	之前获取的权限
---------------	---------

返回

成功返回true, 失败返回false

5.11.3.89 E.SetExternalTCPByID()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.SetExternalTCPByID (
    System::Int32 ecpid )
```

设置当前外部坐标系.

参数

<i>ecpid</i>	外部坐标系变量id
--------------	-----------

返回

成功返回true, 失败返回false

5.11.3.90 E.SetExternalTCPByName()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.SetExternalTCPByName (
    System::String^ ecpName )
```

设置当前外部坐标系.

参数

<i>ecpName</i>	外部坐标系变量名, 例: t.g.EXTTCP1
----------------	--------------------------

返回

成功返回true，失败返回false

5.11.3.91 E.SetFine()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.SetFine (
    System::Double value )
```

设置默认到位精度值，掉电保存.

参数

<i>value</i>	到位精度值
--------------	-------

返回

成功返回true，失败返回false

5.11.3.92 E.SetGlobalSpeed()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.SetGlobalSpeed (
    System::Int32 value )
```

设置全局速度.

参数

<i>value</i>	速度值[0-100]
--------------	------------

返回

成功返回true，失败返回false

5.11.3.93 E.SetIOValue()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.SetIOValue (
    System::Int32 IoIndex,
    E.IOType_CLI IoType,
    System::Collections::Generic::List< System::Double >^ dataIn )
```

设置IO的值.

参数

<i>loIndex</i>	开始的IO下标
<i>loType</i>	IO类型
<i>dataIn</i>	要修改的IO值数组

返回

成功返回true，失败返回false

5.11.3.94 E_SetJogCoord()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_SetJogCoord (
    E_CoordType_CLI coordType )
```

设置点动坐标系类型.

参数

<i>coordType</i>	坐标系类型
------------------	-------

返回

成功返回true，失败返回false

5.11.3.95 E_SetJogEnable()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_SetJogEnable (
    System::Boolean isEnabled )
```

设置当前模式下jog是否可用.

参数

<i>isEnabled</i>	为true，jog可用，为false，jog不可用
------------------	---------------------------

返回

成功返回true，失败返回false

5.11.3.96 E.SetMultiDO()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.SetMultiDO (
    System::Collections::Generic::List< System::Int32 >^ IoValues )
```

设置DO的值.

参数

<i>IoValues</i>	IO值数组, +/-代表1/0, 数值代表DO下标, 例: {-1,2,6,-8}, 表示将DO1、DO8设置为0, DO2、DO6设置为1
-----------------	--

返回

成功返回true, 失败返回false

5.11.3.97 E.SetPosArrayValue()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.SetPosArrayValue (
    E.VarType_CLI varType,
    System::String^ varName,
    E.ScopeType_CLI varScope,
    System::Int32 startIndex,
    System::Int32 nums,
    cli::array< E.ROB.POS_CLI^>^ dataIn )
```

设置点位数组.

参数

<i>varType</i>	变量类型 只可以是CposArray或AposArray,其他类型无效
<i>varName</i>	变量名
<i>varScope</i>	变量域
<i>startIndex</i>	指定存储的起始下标
<i>nums</i>	指定存储的个数
<i>dataIn</i>	存储的点位数组

返回

成功返回true, 失败返回false

5.11.3.98 E.SetRunMode()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.SetRunMode (
    E.RunModeType_CLI mode )
```

设置运动模式.

参数

<i>mode</i>	运动模式
-------------	------

返回

成功返回true，失败返回false

5.11.3.99 E.SetServoOn()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.SetServoOn (
    System::Boolean isOn )
```

设置机器人伺服使能.

参数

<i>isOn</i>	为true上使能，为false掉使能
-------------	--------------------

返回

成功返回true，失败返回false

5.11.3.100 E.SetSysMode()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E.SetSysMode (
    E.SysModeType_CLI sysMode )
```

设置系统模式.

参数

<i>sysMode</i>	系统模式，目前支支持手动和API，其余设置无效
----------------	-------------------------

返回

成功返回true，失败返回false

5.11.3.101 E_SetVarValue()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_SetVarValue (
    E_VarType_CLI varType,
    System::String^ varName,
    System::String^ varValue,
    E_ScopeType_CLI varScope )
```

设置变量.

参数

<i>varType</i>	变量类型
<i>varName</i>	变量名
<i>varValue</i>	变量值, 字符串类型, 需要从其他类型转换
<i>varScope</i>	变量域

返回

成功返回true, 失败返回false

5.11.3.102 E_TeachPoint()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_TeachPoint (
    System::String^ pointName,
    E_ScopeType_CLI scope )
```

示教点.

参数

<i>pointName</i>	位置点变量名
<i>scope</i>	变量域, 不能是系统域

返回

成功返回true, 失败返回false

5.11.3.103 E_UnLoadProgram()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_UnLoadProgram ( )
```

卸载程序.

返回

成功返回true，失败返回false

5.11.3.104 E_UnLoadProject()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_UnLoadProject ( )
```

卸载工程.

返回

成功返回true，失败返回false

5.11.3.105 E_WriteInt()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_WriteInt (
    System::String^ varName,
    System::Int32 varScope,
    System::Int32 value )
```

写入Int型变量值.

参数

<i>varName</i>	变量名称
<i>varScope</i>	变量域
<i>value</i>	变量值

返回

成功返回true，失败返回false

5.11.3.106 E_WritePos()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_WritePos (
    System::String^ varName,
    System::Int32 varScope,
    E_ROB_POS_CLI^ value )
```

写入位置类型变量值.

参数

<i>varName</i>	变量名称
<i>varScope</i>	变量域
<i>value</i>	变量值

返回

成功返回true，失败返回false

5.11.3.107 E_WriteReal()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_WriteReal (
    System::String^ varName,
    System::Int32 varScope,
    System::Double value )
```

写入Double型变量值.

参数

<i>varName</i>	变量名称
<i>varScope</i>	变量域
<i>value</i>	变量值

返回

成功返回true，失败返回false

5.11.3.108 E_WriteStr()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::E_WriteStr (
    System::String^ varName,
    System::Int32 varScope,
    System::String^ value )
```

写入String型变量值.

参数

<i>varName</i>	变量名称
<i>varScope</i>	变量域
<i>value</i>	变量值

返回

成功返回true，失败返回false

5.11.3.109 getLastErr()

```
System::String ^ EstunApiForPublic::EstunApi_CLI::getLastErr ( )
```

获取最近一次接口调用错误的信息.

返回

错误信息字符串

5.11.3.110 isConnected()

```
System::Boolean EstunApiForPublic::EstunApi_CLI::isConnected ( )
```

检测是否已连接服务.

返回

连接返回true，断开返回false

5.11.3.111 setLogConfig()

```
System::Void EstunApiForPublic::EstunApi_CLI::setLogConfig (
    System::String^ logDir,
    System::Int32 size )
```

设置本地日志配置.

参数

<i>logDir</i>	日志文件目录，不指定则在当前目录生成
<i>size</i>	日志文件大小

5.11.3.112 setLogEnable()

```
System::Void EstunApiForPublic::EstunApi_CLI::setLogEnable (
```

```
System::Boolean enable )
```

是否输出本地日志.

参数

<i>enable</i>	为true写本地日志, false则不写日志
---------------	------------------------

5.11.3.113 writeLog()

```
System::Void EstunApiForPublic::EstunApi_CLI::writeLog (
    System::String^ str )
```

写本地日志.

参数

<i>str</i>	日志信息
------------	------

该类的文档由以下文件生成:

- [EstunApiForPublic.h](#)

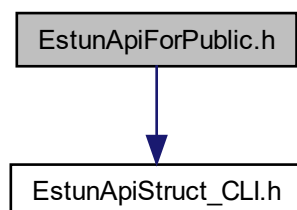
Chapter 6

文件说明

6.1 EstunApiForPublic.h 文件参考

```
#include "EstunApiStruct_CLI.h"
```

EstunApiForPublic.h 的引用(Include)关系图:



类

- class [EstunApiForPublic::EstunApi_CLI](#)

命名空间

- namespace [EstunApiForPublic](#)

6.2 EstunApiForPublic.h

[浏览该文件的文档.](#)

```

00001 #pragma once
00002
00003 #include "EstunApiStruct.CLI.h"
00004
00005 using namespace EstunApiStruct.CLI;
00006
00007 namespace EstunApiForPublic {
00008     public ref class EstunApi.CLI
00009     {
00010     public:
00016         EstunApi.CLI([System::Runtime::InteropServices::Optional] System::String^ configPath);
00017         ~EstunApi.CLI();
00018
00025         System::Void setLogConfig(System::String^ logDir, System::Int32 size);
00026
00032         System::Void setLogEnable(System::Boolean enable);
00033
00039         System::Void writeLog(System::String^ str);
00040
00046         System::String^ getLastErr();
00047
00055         System::Boolean connectToServer(System::String^ ipAddr, System::Boolean isAutoReConnect);
00056
00062         System::Boolean disconnect();
00063
00069         System::Boolean isConnected();
00070
00076         System::Int32 E_GetAPIStatus();
00077
00083         E_ROB_PERMIT.CLI^ E_AcquirePermit();
00084
00090         E_ROB_PERMIT.CLI^ E_CurrentPermit();
00091
00098         System::Boolean E_ReleasePermit(E_ROB_PERMIT.CLI^ permit);
00099
00105         System::Boolean E_MotionStart();
00106
00112         System::Boolean E_MotionPause();
00113
00119         System::Boolean E_MotionContinue();
00120
00126         System::Boolean E_MotionStop();
00127
00133         System::Int32 E_GetRobotRunStatus();
00134
00141         E_ROB_APIBUFF.CLI^ ADDDO.IO.2.E_ROB_APIBUFF(E_ROB_ADDDO.IO.CLI^ from);
00142
00158         E_ROB_JOBID.CLI^ E_MovJ(System::String^ Pname, [System::Runtime::InteropServices::Optional]
System::String^ velName, [System::Runtime::InteropServices::Optional] System::String^ blendType,
[System::Runtime::InteropServices::Optional] System::String^ zname,
[System::Runtime::InteropServices::Optional] System::String^ tname,
[System::Runtime::InteropServices::Optional] System::String^ coordname,
[System::Runtime::InteropServices::Optional] System::String^ payloadName,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF.CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
00159
00174         E_ROB_JOBID.CLI^ E_MovJ2(E_ROB_POS.CLI^ dstPoint, [System::Runtime::InteropServices::Optional]
E_ROB_SPEED.CLI^ velo, [System::Runtime::InteropServices::Optional] E_ROB_ZONE.CLI^ zone,
[System::Runtime::InteropServices::Optional] System::Int32 payloadId,
[System::Runtime::InteropServices::Optional] System::Int32 toolId,
[System::Runtime::InteropServices::Optional] System::Int32 coordId,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF.CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
00175
00192         E_ROB_JOBID.CLI^ E_MovL(System::String^ Pname, [System::Runtime::InteropServices::Optional]
System::String^ velName, [System::Runtime::InteropServices::Optional] System::String^ blendType,
[System::Runtime::InteropServices::Optional] System::String^ zname,
[System::Runtime::InteropServices::Optional] System::String^ tname,
[System::Runtime::InteropServices::Optional] System::String^ coordname,
[System::Runtime::InteropServices::Optional] System::String^ payloadName,
[System::Runtime::InteropServices::Optional] System::String^ overrideType,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF.CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
00193
00209         E_ROB_JOBID.CLI^ E_MovL2(E_ROB_POS.CLI^ dstPoint, [System::Runtime::InteropServices::Optional]
E_ROB_SPEED.CLI^ velo, [System::Runtime::InteropServices::Optional] E_ROB_ZONE.CLI^ zone,
[System::Runtime::InteropServices::Optional] System::Int32 payloadId,
[System::Runtime::InteropServices::Optional] System::Int32 toolId,
[System::Runtime::InteropServices::Optional] System::Int32 coordId,

```



```

[System::Runtime::InteropServices::Optional] System::Int32 overrideType,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
00210
00228 E_ROB_JOBID_CLI^ E_MovC(System::String^ PAuxname, System::String^ PDstname,
[System::Runtime::InteropServices::Optional] System::String^ velName,
[System::Runtime::InteropServices::Optional] System::String^ blendType,
[System::Runtime::InteropServices::Optional] System::String^ zname,
[System::Runtime::InteropServices::Optional] System::String^ tname,
[System::Runtime::InteropServices::Optional] System::String^ coorname,
[System::Runtime::InteropServices::Optional] System::String^ payloadName,
[System::Runtime::InteropServices::Optional] System::String^ overrideType,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
00229
00246 E_ROB_JOBID_CLI^ E_MovC2(E_ROB_POS_CLI^ auxP, E_ROB_POS_CLI^ dstPoint,
[System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
[System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
[System::Runtime::InteropServices::Optional] System::Int32 payloadId,
[System::Runtime::InteropServices::Optional] System::Int32 toolId,
[System::Runtime::InteropServices::Optional] System::Int32 coordId,
[System::Runtime::InteropServices::Optional] System::Int32 overrideType,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
00247
00265 E_ROB_JOBID_CLI^ E_MovArch(System::String^ Pname, [System::Runtime::InteropServices::Optional]
System::String^ velName, [System::Runtime::InteropServices::Optional] System::String^ blendType,
[System::Runtime::InteropServices::Optional] System::String^ zname,
[System::Runtime::InteropServices::Optional] System::String^ tname,
[System::Runtime::InteropServices::Optional] System::String^ coorname,
[System::Runtime::InteropServices::Optional] System::String^ payloadName,
[System::Runtime::InteropServices::Optional] System::String^ cornerName,
[System::Runtime::InteropServices::Optional] System::Double height,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
00266
00283 E_ROB_JOBID_CLI^ E_MovArch2(E_ROB_POS_CLI^ dstPoint, [System::Runtime::InteropServices::Optional]
E_ROB_SPEED_CLI^ velo, [System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
[System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ connerZone,
[System::Runtime::InteropServices::Optional] System::Int32 payloadId,
[System::Runtime::InteropServices::Optional] System::Int32 toolId,
[System::Runtime::InteropServices::Optional] System::Int32 coordId,
[System::Runtime::InteropServices::Optional] System::Double height,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
00284
00302 E_ROB_JOBID_CLI^ E_MovJArch2(E_ROB_POS_CLI^ dstPoint,
[System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
[System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
[System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ connerZone,
[System::Runtime::InteropServices::Optional] System::Int32 payloadId,
[System::Runtime::InteropServices::Optional] System::Int32 toolId,
[System::Runtime::InteropServices::Optional] System::Int32 coordId,
[System::Runtime::InteropServices::Optional] System::Double Upheight,
[System::Runtime::InteropServices::Optional] System::Double Downheight,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
00303
00321 E_ROB_JOBID_CLI^ E_MovLArch2(E_ROB_POS_CLI^ dstPoint,
[System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
[System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
[System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ connerZone,
[System::Runtime::InteropServices::Optional] System::Int32 payloadId,
[System::Runtime::InteropServices::Optional] System::Int32 toolId,
[System::Runtime::InteropServices::Optional] System::Int32 coordId,
[System::Runtime::InteropServices::Optional] System::Double Upheight,
[System::Runtime::InteropServices::Optional] System::Double Downheight,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
00322
00330 System::Boolean E_GetToolsID(System::Int32% num,
System::Collections::Generic::List<System::Int32>% idArray);
00331
00340 System::Boolean E_CreateTool(System::Int32 toolID, E_ROB_COORD_CLI^ toolCoord,
E_ROB_PAYLOAD_CLI^ payloadvalue);
00341
00350 System::Boolean E_ModifyTool(System::Int32 toolID, E_ROB_COORD_CLI^ toolCoord,
E_ROB_PAYLOAD_CLI^ payloadvalue);
00351
00358 System::Boolean E_LoadTool(System::Int32 toolID);

```

```

00359
00365      System::Int32 E_GetCurToolID();
00366
00373      System::String^ E_GetToolName(System::Int32 toolID);
00374
00383      System::Boolean E_GetTool(System::Int32 toolID, E_ROB_COORD_CLI^% toolCoord, E_ROB_PAYLOAD_CLI^%
toolPayload);
00384
00392      System::Boolean E_GetUserCoordsID(System::Int32% num,
System::Collections::Generic::List<System::Int32>^% idArray);
00393
00401      System::Boolean E_CreateUserCoord(System::Int32 usrID, E_ROB_COORD_CLI^ usrData);
00402
00410      System::Boolean E_ModifyUserCoord(System::Int32 usrID, E_ROB_COORD_CLI^ usrData);
00411
00418      System::Boolean E_LoadUserCoord(System::Int32 usrID);
00419
00426      System::Int32 E_GetCurUserCoordID(E_UserCoordType_CLI% coordtype);
00427
00435      System::String^ E_GetUserCoordName(System::Int32 coordID, E_UserCoordType_CLI coordtype);
00436
00445      System::Boolean E_GetUserCoord(System::Int32 coordID, E_UserCoordType_CLI coordtype,
E_ROB_COORD_CLI^% coord);
00446
00454      System::Boolean E_GetPayLoadsID(System::Int32% num,
System::Collections::Generic::List<System::Int32>^% idArray);
00455
00463      System::Boolean E_CreatePayload(System::Int32 payloadID, E_ROB_PAYLOAD_CLI^ payloadData);
00464
00472      System::Boolean E_ModifyPayload(System::Int32 payloadID, E_ROB_PAYLOAD_CLI^ payloadData);
00473
00480      System::Boolean E_LoadPayload(System::Int32 payloadID);
00481
00487      System::Int32 E_GetCurPayloadID();
00488
00495      System::String^ E_GetPayloadName(System::Int32 payloadID);
00496
00504      System::Boolean E_GetPayload(System::Int32 payloadID, E_ROB_PAYLOAD_CLI^% payload);
00505
00514      System::Boolean E_GetDefaultPayload(System::Int32% payloadID, System::String^% name,
E_ROB_PAYLOAD_CLI^% payload);
00515
00524      System::Boolean E_SetIOValue(System::Int32 IoIndex, E_IOTYPE_CLI IoType,
System::Collections::Generic::List<System::Double>^% dataIn);
00525
00535      System::Boolean E_GetIOValue(System::Int32 IoIndex, E_IOTYPE_CLI IoType, System::Int32 IoNum,
System::Collections::Generic::List<System::Double>^% dataOut);
00536
00546      System::Boolean E_SetVarValue(E_VarType_CLI varType, System::String^ varName, System::String^
varValue, E_ScopeType_CLI varScope);
00547
00557      System::Boolean E_GetVarValue(E_VarType_CLI varType, System::String^ varName, E_ScopeType_CLI
varScope, System::String^% valueBack);
00558
00565      System::Boolean E_SetFine(System::Double value);
00566
00573      System::Boolean E_GetFine(System::Double % value);
00574
00581      System::Boolean E_SetJogCoord(E_CoordType_CLI coordType);
00582
00588      E_CoordType_CLI E_GetJogCoord();
00589
00596      System::Boolean E_SetJogEnable(System::Boolean isEnabled);
00597
00603      System::Boolean E_GetJogEnable();
00604
00613      System::Boolean E_JogMotion(System::Int32 axisId, E_DistentionType_CLI distention,
System::Boolean mov);
00614
00624      System::Boolean E_JogMicroMotion(System::Int32 axisId, E_DistentionType_CLI distention,
System::Double dist, System::Boolean mov);
00625
00633      System::Boolean E_TeachPoint(System::String^ pointName, E_ScopeType_CLI scope);
00634
00641      System::Boolean E_GetRobotType(System::String^% value);
00642
00649      System::Boolean E_GetRobotInfo(System::String^% value);
00650
00656      System::Int32 E_GetRobotStatus();
00657
00664      System::Boolean E_SetGlobalSpeed(System::Int32 value);
00665
00671      System::Int32 E_GetGlobalSpeed();
00672
00679      System::Boolean E_SetSysMode(E_SysModeType_CLI sysMode);
00680
00686      E_SysModeType_CLI E_GetSysMode();

```

```

00687
00694     System::Boolean E_SetRunMode(E_RunModeType_CLI mode);
00695
00701     E_RunModeType_CLI E_GetCurRunMode();
00702
00709     System::Boolean E_SetServoOn(System::Boolean isOn);
00710
00716     E_ServoStatusType_CLI E_GetServoOn();
00717
00723     System::Boolean E_CheckClearError();
00724
00730     System::Boolean E_ClearError();
00731
00737     System::Int32 E_GetErrorId();
00738
00745     System::String^ E_GetErrorString(System::Int32 errorId);
00746
00754     System::String^ E_GetErrorString(System::Int32 errorId, System::String^ language);
00755
00761     E_ROB_POS_CLI^ E_GetCurJPos();
00762
00768     E_ROB_POS_CLI^ E_GetCurWPos();
00769
00776     System::Boolean E_SetExternalTCPByName(System::String^ ecpname);
00777
00784     System::Boolean E_SetExternalTCPByID(System::Int32 ecpid);
00785
00792     System::Boolean E_GetCurProgramName(E_ROB_PROGNAME_CLI^ % value);
00793
00799     E_RunStatusType_CLI E_GetCurProgramRunStatus();
00800
00806     System::Boolean E_UnLoadProject();
00807
00813     System::Boolean E_UnLoadProgram();
00814
00821     System::Boolean E_LoadProject(System::String^ prjname);
00822
00830     System::Boolean E_LoadProgame(System::String^ prjname, System::String^ programname);
00831
00837     System::Boolean E_ProgRun();
00838
00844     System::Boolean E_ProgPause();
00845
00851     System::Boolean E_ProgStop();
00852
00861     System::Boolean E_ReadInt(System::String^ varName, System::Int32 varScope, System::Int32 %
valueBack);
00862
00871     System::Boolean E_WriteInt(System::String^ varName, System::Int32 varScope, System::Int32
value);
00872
00881     System::Boolean E_ReadReal(System::String^ varName, System::Int32 varScope, System::Double %
valueBack);
00882
00891     System::Boolean E_WriteReal(System::String^ varName, System::Int32 varScope, System::Double
value);
00892
00901     System::Boolean E_ReadStr(System::String^ varName, System::Int32 varScope, System::String^%
valueBack);
00902
00911     System::Boolean E_WriteStr(System::String^ varName, System::Int32 varScope, System::String^
value);
00912
00921     System::Boolean E_ReadPos(System::String^ varName, System::Int32 varScope, E_ROB_POS_CLI^%
valueBack);
00922
00931     System::Boolean E_WritePos(System::String^ varName, System::Int32 varScope, E_ROB_POS_CLI^
value);
00932
00939     System::Boolean E_SetMultiDO(System::Collections::Generic::List<System::Int32>^ IoValues);
00940
00946     E_ROB_JOBID_CLI^ E_GetCurMovCmdJobId();
00947
00953     System::Int32 E_GetCurMovCmdSts();
00954
00961     System::Int32 E_GetMovCmdSts(E_ROB_JOBID_CLI^ jobId);
00962
00974     System::Boolean E_SetPosArrayValue(E_VarType_CLI varType, System::String^ varName,
E_ScopeType_CLI varScope, System::Int32 startIndex, System::Int32 nums, cli::array<E_ROB_POS_CLI>^%
dataIn);
00975
00987     System::Boolean E_GetPosArrayValue(E_VarType_CLI varType, System::String^ varName,
E_ScopeType_CLI varScope, System::Int32 startIndex, System::Int32 nums, cli::array<E_ROB_POS_CLI>^%
dataOut);
00988
00995     System::Boolean E_GetAxeTorqueValue(System::Collections::Generic::List<System::Double>^%
dataOut);

```

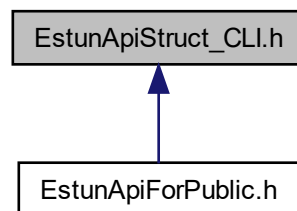
```

00996
01012     E_ROB_JOBID_CLI^ E_MovJ2_Trajectory(cli::array<E_ROB_POS_CLI^>^ dstPointArray, System::Int32
posNums, [System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
[System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
[System::Runtime::InteropServices::Optional] System::Int32 payloadId,
[System::Runtime::InteropServices::Optional] System::Int32 toolId,
[System::Runtime::InteropServices::Optional] System::Int32 coordId,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
01013
01030     E_ROB_JOBID_CLI^ E_MovL2_Trajectory(cli::array<E_ROB_POS_CLI^>^ dstPointArray, int posNums,
[System::Runtime::InteropServices::Optional] E_ROB_SPEED_CLI^ velo,
[System::Runtime::InteropServices::Optional] E_ROB_ZONE_CLI^ zone,
[System::Runtime::InteropServices::Optional] System::Int32 payloadId,
[System::Runtime::InteropServices::Optional] System::Int32 toolId,
[System::Runtime::InteropServices::Optional] System::Int32 coordId,
[System::Runtime::InteropServices::Optional] System::Int32 overrideType,
[System::Runtime::InteropServices::Optional] E_ROB_APIBUFF_CLI^ addDo,
[System::Runtime::InteropServices::Optional] System::Double valfine,
[System::Runtime::InteropServices::Optional] System::Boolean IsWaitFinish);
01031
01038     System::Int32 E_GetTrajectoryCmdSts(E_ROB_JOBID_CLI^ posArrayJobID);
01039
01048     System::Boolean E_GetTrajectoryPointsJobId(E_ROB_JOBID_CLI^ posArrayJobID,
cli::array<E_ROB_JOBID_CLI^>^ posAllJobIDBuff, int buffNums);
01049
01057     System::Boolean E_CposToApos(System::Collections::Generic::List<System::Double>^ Cpos,
System::Collections::Generic::List<System::Double>^ Apos);
01058
01066     System::Boolean E_AposToCpos(System::Collections::Generic::List<System::Double>^ Apos,
System::Collections::Generic::List<System::Double>^ Cpos);
01067
01073     System::Int32 E_GetCurTcpVelo();
01074 };
01075 }

```

6.3 EstunApiStruct_CLI.h 文件参考

此图展示该文件直接或间接的被哪些文件引用了:



类

- struct [EstunApiStruct_CLI::E_ROB_PERMIT_CLI](#)
- struct [EstunApiStruct_CLI::E_ROB_JOBID_CLI](#)
- struct [EstunApiStruct_CLI::E_ROB_PROGNAME_CLI](#)
- struct [EstunApiStruct_CLI::E_ROB_APIBUFF_CLI](#)
- struct [EstunApiStruct_CLI::E_ROB_ADDDO_IO_CLI](#)
- struct [EstunApiStruct_CLI::E_ROB_POS_CLI](#)
- struct [EstunApiStruct_CLI::E_ROB_SPEED_CLI](#)
- struct [EstunApiStruct_CLI::E_ROB_ZONE_CLI](#)
- struct [EstunApiStruct_CLI::E_ROB_PAYLOAD_CLI](#)
- struct [EstunApiStruct_CLI::E_ROB_COORD_CLI](#)

命名空间

- namespace [EstunApiStruct_CLI](#)

宏定义

- #define [APIBUFFLENGTH](#) 512

枚举

- enum class [EstunApiStruct_CLI::E_IOType_CLI](#) {
[EstunApiStruct_CLI::DI](#) = 0 , [EstunApiStruct_CLI::DO](#) = 1 , [EstunApiStruct_CLI::AO](#) = 2 , [EstunApiStruct_CLI::AI](#) = 3 ,
[EstunApiStruct_CLI::SIM_DI](#) = 11 , [EstunApiStruct_CLI::SIM_DO](#) = 12 , [EstunApiStruct_CLI::SIM_AI](#) = 13 ,
[EstunApiStruct_CLI::SIM_AO](#) = 14 }
- enum class [EstunApiStruct_CLI::E_VarType_CLI](#) {
[EstunApiStruct_CLI::OtherType](#) = 0 , [EstunApiStruct_CLI::intType](#) = 1 , [EstunApiStruct_CLI::realType](#) = 2 ,
[EstunApiStruct_CLI::aposType](#) = 3 ,
[EstunApiStruct_CLI::cpoSType](#) = 4 , [EstunApiStruct_CLI::strType](#) = 5 , [EstunApiStruct_CLI::RealIntArray](#) = 6 ,
[EstunApiStruct_CLI::tooType](#) = 7 ,
[EstunApiStruct_CLI::userType](#) = 8 , [EstunApiStruct_CLI::CposArray](#) = 11 , [EstunApiStruct_CLI::AposArray](#) = 12 }
- enum class [EstunApiStruct_CLI::E_ScopeType_CLI](#) { [EstunApiStruct_CLI::system](#) = 0 , [EstunApiStruct_CLI::global](#) ,
[EstunApiStruct_CLI::project](#) , [EstunApiStruct_CLI::local](#) }
- enum class [EstunApiStruct_CLI::E_CoordType_CLI](#) {
[EstunApiStruct_CLI::axisCoord](#) = 0 , [EstunApiStruct_CLI::worldCoord](#) , [EstunApiStruct_CLI::toolCoord](#) ,
[EstunApiStruct_CLI::usrCoord](#) ,
[EstunApiStruct_CLI::errCoord](#) }
- enum class [EstunApiStruct_CLI::E_UserCoordType_CLI](#) { [EstunApiStruct_CLI::UsrCoord](#) = 0 , [EstunApiStruct_CLI::ExtCoord](#) }
- enum class [EstunApiStruct_CLI::E_DistentionType_CLI](#) { [EstunApiStruct_CLI::negative](#) = 0 , [EstunApiStruct_CLI::forward](#) }
- enum class [EstunApiStruct_CLI::E_SysModeType_CLI](#) {
[EstunApiStruct_CLI::manualMode](#) = 0 , [EstunApiStruct_CLI::autoMode](#) , [EstunApiStruct_CLI::remoteMode](#) ,
[EstunApiStruct_CLI::banMode](#) ,
[EstunApiStruct_CLI::mutil](#) , [EstunApiStruct_CLI::API](#) , [EstunApiStruct_CLI::errSysMode](#) }
- enum class [EstunApiStruct_CLI::E_RunModeType_CLI](#) { [EstunApiStruct_CLI::step](#) = 1 , [EstunApiStruct_CLI::sustain](#) ,
[EstunApiStruct_CLI::errRunMode](#) }
- enum class [EstunApiStruct_CLI::E_RunStatusType_CLI](#) { [EstunApiStruct_CLI::running](#) = 1 , [EstunApiStruct_CLI::suspend](#) ,
[EstunApiStruct_CLI::stop](#) , [EstunApiStruct_CLI::errRunStatus](#) }
- enum class [EstunApiStruct_CLI::E_ServoStatusType_CLI](#) { [EstunApiStruct_CLI::ServoOff](#) = 0 , [EstunApiStruct_CLI::ServoOn](#) ,
[EstunApiStruct_CLI::errStatus](#) }

6.3.1 宏定义说明

6.3.1.1 APIBUFFLENGTH

```
#define APIBUFFLENGTH 512
```

在文件 [EstunApiStruct_CLI.h](#) 第 3 行定义.

6.4 EstunApiStruct_CLI.h

[浏览该文件的文档.](#)

```

00001 #pragma once
00002
00003 #define APIBUFFLENGTH 512
00004
00005 namespace EstunApiStruct_CLI
00006 {
00011     public enum class E_IOType_CLI
00012     {
00013         DI = 0,
00014         DO = 1,
00015         AO = 2,
00016         AI = 3,
00017         SIM_DI = 11,
00018         SIM_DO = 12,
00019         SIM_AI = 13,
00020         SIM_AO = 14,
00021     };
00022
00027     public enum class E_VarType_CLI
00028     {
00029         OtherType = 0,
00030         intType = 1,
00031         realType = 2,
00032         aposType = 3,
00033         cposType = 4,
00034         strType = 5,
00035         RealIntArray = 6,
00036         tooType = 7,
00037         userType = 8,
00038         CposArray = 11,
00039         AposArray = 12
00040     };
00041
00046     public enum class E_ScopeType_CLI
00047     {
00048         system = 0,
00049         global,
00050         project,
00051         local,
00052     };
00053
00058     public enum class E_CoordType_CLI
00059     {
00060         axisCoord = 0,
00061         worldCoord,
00062         toolCoord,
00063         usrCoord,
00064         errCoord,
00065     };
00066
00071     public enum class E_UserCoordType_CLI
00072     {
00073         UsrCoord = 0,
00074         ExtCoord,
00075     };
00076
00081     public enum class E_DistentionType_CLI
00082     {
00083         negative = 0,
00084         forward,
00085     };
00086
00091     public enum class E_SysModeType_CLI
00092     {
00093         manualMode = 0,
00094         autoMode,
00095         remoteMode,
00096         banMode,
00097         mutil,
00098         API,
00099         errSysMode,
00100     };
00101
00106     public enum class E_RunModeType_CLI
00107     {
00108         step = 1,
00109         sustain,
00110         errRunMode,
00111     };
00112
00117     public enum class E_RunStatusType_CLI
00118     {

```

```

00119         running = 1,
00120         suspend,
00121         stop,
00122         errRunStatus,
00123     };
00124
00129     public enum class E_ServoStatusType_CLI
00130     {
00131         ServoOff = 0,
00132         ServoOn,
00133         errStatus,
00134     };
00135
00139     public ref struct E_ROB_PERMIT_CLI
00140     {
00144         System::UInt16 m_mainctrlcode;
00148         System::UInt16 m_timestamp;
00149     };
00150
00154     public ref struct E_ROB_JOBID_CLI
00155     {
00159         System::Int32 m_jobID;
00163         System::UInt32 m_timeStamp;
00164     };
00165
00169     public ref struct E_ROB_PROGNAME_CLI
00170     {
00174         System::String^ prjname;
00178         System::String^ programname;
00182         System::String^ exeprogramName;
00183     };
00184
00188     public ref struct E_ROB_APIBUFF_CLI
00189     {
00194         E_ROB_APIBUFF_CLI()
00195         {
00196             m_type = 0x00;
00197             m_data = gcnew System::Collections::Generic::List<System::SByte>();
00198         }
00202         System::SByte m_type;
00206         System::Collections::Generic::List<System::SByte>^ m_data;
00207     };
00208
00212     public ref struct E_ROB_ADDDO_IO_CLI
00213     {
00217         E_IOTType_CLI ioType;
00221         System::Int32 ioindex;
00225         System::Double iovalue;
00229         System::Int32 trigType;
00233         System::Int32 exTimes;
00237         System::Int32 exPercent;
00241         System::Double exDistance;
00242     };
00243
00247     public ref struct E_ROB_POS_CLI
00248     {
00253         E_ROB_POS_CLI()
00254         {
00255             posCfg = gcnew System::Collections::Generic::List<System::Int32>();
00256             posValue = gcnew System::Collections::Generic::List<System::Double>();
00257         }
00258
00262         System::Int32 posType;
00266         System::Collections::Generic::List<System::Int32>^ posCfg;
00270         System::Collections::Generic::List<System::Double>^ posValue;
00271     };
00272
00276     public ref struct E_ROB_SPEED_CLI
00277     {
00281         System::Double m_Per;
00285         System::Double m_Tcp;
00289         System::Double m_ori;
00293         System::Double m_exj-l;
00297         System::Double m_exj-r;
00298     };
00299
00303     public ref struct E_ROB_ZONE_CLI
00304     {
00308         System::Int32 m_ZoneType;
00312         System::Double m_Per;
00316         System::Double m_dis;
00320         System::Int32 m_vConst;
00321     };
00322
00326     public ref struct E_ROB_PAYLOAD_CLI
00327     {
00331         System::Double m_M;

```

```
00335         System::Double m_Mx;
00339         System::Double m_My;
00343         System::Double m_Mz;
00347         System::Double m_Ixx;
00351         System::Double m_Iyy;
00355         System::Double m_Izz;
00359         System::Double m_Ixy;
00363         System::Double m_Ixz;
00367         System::Double m_Iyz;
00368     };
00369
00373     public ref struct E_ROB.COORD.CLI
00374     {
00378         System::Double x;
00382         System::Double y;
00386         System::Double z;
00390         System::Double a;
00394         System::Double b;
00398         System::Double c;
00399     };
00400 }
```


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